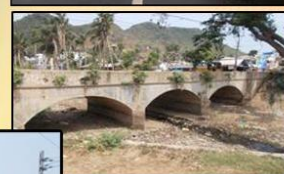


ENVIRONMENTAL ASSESSMENT REPORT

**Consultancy Services for
Preparation of Detailed Project Reports for
Visakhapatnam – Chennai Industrial Corridor
Connectivity Project (Package - 1)**

Agreement No.01/CE(R&B)CRN, PPP & MD, APRDC/ADB/2015-16



ANAKAPALLE - ATCHUTAPURAM **Volume VI : Environmental Assessment**

MARCH 2016

Prepared by



Roughton International Ltd

in Joint Venture with



SATRA Infrastructure Management Services Pvt. Ltd

Prepared for



**Andhra Pradesh Road
Development Corporation**



**GOVERNMENT OF ANDHRA PRADESH
ROADS AND BUILDINGS DEPARTMENT**



**Consultancy Services for preparation of Detailed Project Report
for Visakhapatnam – Chennai Industrial Corridor Connectivity
Project-Package-I (Anakapalle to Atchutapuram Road)**

QUALITY ASSURANCE STATEMENT		
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REPORT NAME: ENVIRONMENTAL ASSESSMENT / INITIAL ENVIRONMENTAL EXAMINATION ANAKAPALLE TO ATCHUTAPURAM		REVIEWED BY: Sankaran.V, Ashik Hussain, Surya Prakash, Venkata Krishna, Prashant Joshi
PROJECT/CONTRACT NUMBER: <u>For APRDC : 01/CE(R&B), CRN, PPP&MD, APRDC/ADB/2015-16, Dt.05-06-2015</u> <u>For Roughton International Ltd. in JV with SATRA Infrastructure Management Services Pvt. Ltd. : 01041004</u>		APPROVED FOR ISSUE BY: Raj Mallela
REVISIONS:		DATE OF ISSUE: March 2016

ANDHRA PRADESH ROAD DEVELOPMENT CORPORATION (APRDC)

Consultancy Services for preparation of Detailed Project Report for Visakhapatnam – Chennai Industrial Corridor Connectivity Project-Package-I (Anakapalle to Atchutapuram Road)

FINAL REPORT VOLUMES

Report Name	Volume Label
Report-Main Report	Volume I
Annexures: Traffic, Highways, Soil and Materials, Pavement, Economic Analysis and Social and Environmental Assessment, Geotechnical Report and Hydrological Calculations of Drainage Structures, Detailed Design Calculations of VUP, Minor Bridges, Culverts and Retaining Wall	Volume II A
Annexures: Detailed Design Calculations of Flyover	Volume II B
Drawings: Highways and Structures	Volume III A
Drawings: Utility Plan	Volume III B
Engineer's Cost Estimate and Rate Analysis	Volume IV
Tender Documents	Volume V
Environmental Assessment	Volume VI
Social Assessment and Resettlement Plan & Annexures	Volume VII
Land Plan Schedules	Volume VIII

ABBREVIATIONS

ADB	–	Asian Development Bank
APRDC	–	Andhra Pradesh Road Development Corporation
DC	–	District Collector
DH	–	Displaced Household
DP	–	Displaced Person
EA	–	Executing Agency
GOI	–	Government of India
GRC	–	Grievance Redressal Committee
IA	–	Implementing Agency
IAY	–	Indira Awaas Yojana
JC	–	Joint Collector
LA	–	Land Acquisition
RFCTLARR	–	The Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013
NGO	–	Nongovernment organization
PD	–	Project Director
PIU	–	Project implementation Unit
PRoW	–	Proposed Right-of-Way
RDO	–	Revenue Divisional Officer
R&R	–	Rehabilitation and Resettlement
RF	–	Resettlement Framework
RO	–	Resettlement Officer
RoW	–	Right-of-Way
RP	–	Resettlement Plan
SC	–	Scheduled Caste
SH	–	State Highway
SMU	–	Social Management Unit
SPS	–	Safeguard Policy Statement
SSR	–	Standard Schedule of Rates
ST	–	Scheduled Tribe
VDA	–	Variable Dearness Allowance

VRO – Village Revenue Officer

DEFINITIONS

Affected Family: Any family affected by the project, living, cultivating in land or carrying on business, trade or any other occupation within the Corridor of Impact (CoI) who is impacted by the project is an Affected Family (AF).

Affected Person: Any individual, part of the affected family living, cultivating in land or carrying on business, trade or any other occupation within the Corridor of Impact (CoI) who is impacted by the project is an Affected Person (AP).

Agricultural Land: Land used for the purpose of: (i) agriculture or horticulture; (ii) dairy farming, poultry farming, pisciculture, sericulture, seed farming; (iii) breeding of livestock or nursery growing medicinal herbs; (iv) raising of crops, trees, grass or garden produce; and (v) land used for the grazing of cattle.

Assigned Land: Are government cultivable waste land provided to the land less poor persons for their lively hood more particularly land less agricultural labour and if the land is required for public purpose, on payment of compensation, the land can be resumed to govt.

Assistance: All support mechanisms such as subsistence allowance, shifting assistance, livelihood assistance, involving monetary help, services or assets given to APs constitute assistance in this project.

Compensation: Compensation refers to amount paid to acquired private land and/or structure in accordance with the provisions of Sec 26, Sec 27, Sec 28, Sec 29 and Sec 30 pertaining to compensation and Sec 31 and Sec 32 pertaining to rehabilitation and resettlement contained in 'The Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013'.

Cut-off Date: For title holders, the date of SIA notification [Sec 4(2)] of intended acquisition as per the provisions of RFCTLARR Act will be treated as the cut-off date, and for non-titleholders the start date of project census survey for the subproject will be the cut-off date.

Displaced Family (DF): Any family living, cultivating land or carrying on business, trade or any other occupation within the Proposed Right-of-Way (PRoW) who are impacted by the project and have to relocate is a Displaced Family. All the members of a DF in the project will be treated as DPs.

Displaced Person (DP): Any individual or part of the DFs living, cultivating land or carrying on business, trade or any other occupation within the PRoW who are impacted by the project is a Displaced Person (DP).

Encroacher: Any person illegally occupying public property by extending their land boundary or a portion of their building onto the RoW is an encroacher.

Entitled Person (EP): Entitled Person includes all APs who qualify for, or are entitled to, compensation / assistance since being impacted by the project. The basis for identification of Entitled Persons (EP) in the project will be the cut-off date.

Marginal Farmer: A cultivator with an un-irrigated land holding up to one hectare or irrigated land holding up to one-half hectare.

Right-of-Way: Right-of-Way (RoW) is the land that have been granting to be owned by PWD(R&B) abetting the carriage way and including the carriageway and recorded as PWD (R&B) land in the revenue records

Proposed Right-of-Way: The proposed right-of-way (PRoW) is the width required for the actual construction of the road including the carriageway, shoulder, embankment, and longitudinal drainage, wayside amenities like bus stops, bus shelters, and etc. and necessary safety zones.

Temporary Building: Temporary building means a temporary type of structure, which includes buildings with roofs constructed of thatch, galvanized iron or asbestos.

Significant Impact: The APs suffering the following impacts and requiring to relocate are categorized as significantly impacted APs: (i) loss of place of dwelling, (ii) loss of place of business; (iii) loss of livelihood; and (iv) loss of agricultural productive land.

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Executive Summary

Project Description: The Government of Andhra Pradesh is planning to take up Industrial Development program in Vizag – Chennai Industrial Corridor (VCIC) with ADB loan assistance. As part of this project, Andhra Pradesh Road Development Corporation/ Roads and Buildings Department have been entrusted with the responsibility to identify and develop certain roads connecting proposed industrial clusters to National Highways, Ports, Airports, Railway Stations, urban centres, etc., which are mostly divided roads with a median.

The Government of Andhra Pradesh has taken up Visakhapatnam - Chennai Industrial Corridor – Development programme (VCIC-DP) for infrastructure development and Institutional reforms to stimulate economic growth and employment generation in ANDHRA PRADESH with ADB loan assistance.

The sub project Anakapalle - Atchutapuram road section has been proposed to implement, upgrade the two lane road to four lane road on priority based. With regards to transport corridor, the following road sub-project was proposed for development under VCIC-DP -Package -I.

Environmental Sensitivity and Project Categorization: Project road is not passing through any wildlife sanctuary, national park, tiger reserve, protected area or any other similar eco-sensitive areas. Few sections of some sub-projects pass through protected or reserved forests. All other impacts are site-specific and can be addressed through proven mitigation measures. Hence, the project is classified as Category B warranting an initial environmental examination (IEE) which has been conducted in consistent to Safeguard Policy Statement (SPS), 2009.

Existing Environment: A brief description about the existing environment, including its physical and ecological resources, economic development of the region, and issues relating to quality of life are presented in this section. Broad aspects on various environmental parameters (geology, soil, topography, climate, land use, water resources, water quality, air quality, noise quality, tourism, cultural resources etc.) which are likely to be affected (direct or indirect) by the proposed road improvement project are covered. These aspects are covered in broader geographic extent to present the entire project region.

Andhra Pradesh is eighth largest state of the country has a geographical area of 1.6 lakh sq. Km, which constitutes 5.05% of the land area of the country. The project area lies between latitude 17°41'23" North & longitude 83°0'8.45" East and latitude 17°33'50" North & longitude 82°58'44.57" East.

The proposed project road between **Anakapalle - Atchutapuram** is situated on deltaic plain of south. The soil type in the area is predominantly red loamy soils. Stratigraphically, the project area mainly comprises Achaeans, Laments, Deccan Trap and Recent formation.

Policy, Legal and Administrative Framework for Environmental Management in India

Government of India has enacted more than thirty environment conservation laws and Acts. Some of these are: The Environment (Protection) Act 1986, The Wildlife (Protection) Act 1972; The Forest (Conservation) Act 1980; The Water (Prevention and Control of Pollution) Act 1974, The Air (Prevention and Control of Pollution) Act 1981; 1988 amendment of The Motor Vehicle Act (M.V.) Act, 1939. The Water and The Air Acts entrusted the task of their implementation and regulation to pollution control boards set up for such purpose at State and Central levels.

Screening of Potential Environmental Impacts and Mitigation Measures

This section presents the key environmental issues associated with different aspects of the proposed project. Identification and assessment of the potential environmental impacts are based on secondary information supplemented by data collected by field visits. Impacts on various environmental components have been assessed at four different stages namely Project location, Design, Construction and Operational stages.

Environmental Management & Monitoring Plan and Institutional Assessment

The management and monitoring mechanism required to ensure that the project implementation is carried out in accordance with the due regards to environment. These includes major environmental issues and associated impacts, suggested mitigation measures, implementation and supervising responsibilities, a monitoring plan and institutional assessment and training requirements for successful implementation of the mitigation measures.

Environmental Management Plan: Sub-project specific Environmental Management Plan (EMP) has been formulated with an intend to set out action required to avoid or mitigate all impacts and the responsibility for taking each action. Responsibility is made legally binding when actions are subsequently specified in contracts. Environmental Monitoring Plan (EMoP) has been prepared to ensure that the intended environmental mitigations are realized and these results in desired benefits to the target population causing minimal deterioration to the environmental parameters. All costs for implementing the mitigation measures and monitoring plan will be included in the Bill of Quantities (BOQ) by the

contractor as implementation of the EMP will be the responsibility of the contractor.

APRDC through its Project Implementation Unit (PIU), will ensure the effective implementation of the environmental management plan. APRDC will be assisted by a Project Management Consultant (PMC). There is a need for the PIU to organize its environmental unit to ensure that contractors maintain environmental safeguard compliance. To provide regular monitoring information and technical advice to the PIU a Construction Supervision Consultant will be engaged who will be responsible to examine environmental compliances and suggest corrective actions.

Grievance Redress Mechanism

All the three parties involved in this project implementation i.e. Contractor, CSC and executing agency will maintain complaint registers at their following respective offices Contractor's main site offices i.e. office of the Project Manager, CSC's main site office i.e. office of the Engineer's Representative.

Public Consultation and Information Disclosure

Extensive consultations were made with local communities and government agencies like Forests and Wildlife, State Pollution Control Board, A.P. Space Application Centre, Economics and Statistics to incorporate their views and suggestions. Local community strongly support the project. They disseminated many important information's and made several suggestions and demands. Main demands include adequate compensation and assistance for loss of land and assets, employment in road construction and petty contract provision of safety measures, side drains in built-up areas, avenue plantation and most of their demand have been integrated in design.

Conclusion

This initial environmental examination (IEE) ascertains that upgrading is unlikely to cause any significant environmental impacts. Few impacts were identified attributable to the proposed subproject, all of which are localized and temporary in nature and can be easily mitigated with minor to negligible residual impacts. Need of undertaking detailed EIA is not envisaged at this stage.

1 Introduction

1.1 General

The Government of Andhra Pradesh is planning to take up Industrial Development Program in Visakhapatnam – Chennai Industrial Corridor (VCIC) with the loan assistance of Asian Development Bank (ADB). As a part of the project, Andhra Pradesh Road Development Corporation/Roads and Buildings Department (GOAP/RBD) has been entrusted with the responsibility to identify and develop certain roads connecting proposed Industrial Clusters to National Highways, Ports, Airports, Railway stations, urban centres, etc., which are mostly divided roads with a median. Andhra Pradesh Road Development Corporation (APRDC) will be the implementing agency for the project, on behalf of RBD.

The project roads are located in the State of Andhra Pradesh (AP) and can be referred as part of coastal corridors. The Phase-I Package falls under three districts of AP namely East Godavari, Visakhapatnam (Vizag) and Nellore.

Kakinada consists of two regions, connected by bridges. The southern part, Jagannathapuram, is separated from the rest of the city by the Buckingham Canal. The canal and its branches form Medline Island, which abuts the city in the southwest. The northern area of Kakinada is an industrial belt, running north-south. The length of the city separates the eastern part from the coast. Kakinada is bordered on the southeast by Kakinada Bay and a marshy wetland.

Visakhapatnam (nicknamed as Vizag and sometimes known by its historical name Waltair) district is situated in the Eastern Ghats and on the coast of Bay of Bengal on the east. The city is known for heavy industries, steel plant, seaport, natural harbour and shipyard.

Visakhapatnam District is one of the North Eastern Coastal districts of Andhra Pradesh and it lies between 17°-15' and 18°-32' Northern latitude and 83°-54' and 83°-30' in Eastern longitude. It is bounded on the North partly by the Orissa State and partly by Vizianagaram District, on the South by East Godavari District, on the West by Orissa State and on the East by Bay of Bengal

The main objectives of the improvements are to improve the regional, as well as inter and intra state transport flows, and in doing so improving access to services, and making the State attractive to developers and investors. To fulfil the above objectives and due considerations to environmental feasibility of above road section, Initial Environmental Examination (IEE) was carried out for this section.

The IEE study was started in the month of July 2015 and continuing. This report has been prepared by DPR consultant on the behalf of Executing agency.

1.2 Purpose

The purpose of this Initial Environmental Examination (IEE) is to document a screening of the environmental consequences of Anakapalle - Atchutapuram Road section of Visakhapatnam District. This IEE is structured in accordance with the requirements of the ADB.

1.3 Scope

The IEE of the proposed road section has been undertaken to identify and to minimise the adverse environmental impacts, if any, associated with construction and operation. Environmental issues considered in this IEE are mainly macro-level and may have adverse or beneficial impacts. Initial Environmental Examination of the roads has four basic objectives, as stated below, to:

- Determine the magnitude of potential environmental concerns and to ensure that environmental considerations are given adequate weight when carrying out the proposed road constructions / improvements;
- Identify the environmental issues that should be taken into account when new links are constructed and/or existing links are improved - such as to include both adverse and beneficial impacts;
- Identify any further need for environmental studies or Environmental Impact Assessment (EIA); and Identify potential enhancement / mitigation measures, if any.

1.4 Executing Agency

The executing agency shall be Andhra Pradesh Road Development Corporation (APRDC), Government of India. The project will be implemented by APRDC through its network at District level. The work of the detailed project preparation has been assigned to "Roughton International Ltd, UK in association with Satra I-MAN Private Ltd." Initial Environmental Examination study has been undertaken by Technical Assistance consultant in coordination with DPR consultants.

1.5 Study & Methodology

The Initial Environmental Examination has been carried out using current ADB and Government of India guidelines, specifically

- "Environmental Assessment Guidelines (ADB 2003)";
- "Environmental Guidelines for Selected Infrastructure Projects (ADB 1993b)";
- Safeguard Policy Statement (ADB 2009);

- "Project Terms of Reference (TOR)";
- "Environmental Guidelines for Road/Rail/Highway Projects", Government of India, 1989;
- "Handbook of Environmental Procedures and Guidelines", Government of India 1994;
- "Guidelines for Environmental Impact Assessment of Highway Projects" (IRC: 104-1988); and
- The Environmental (Protection) Act, 1986

The IEE was carried out in four stages:

- Reviews of literature, laws, guidelines and meetings with concerned agencies and organisations, State Authorities (Andhra Pradesh Pollution Control Board, Archaeological Department, District Forest Department, State Ground Water Board, APRDC) in Hyderabad and on-site;
- Reconnaissance survey along with local people consultation as part of information collection was undertaken (in the month of July & August 2015) and process of public consultation organised (on 10.10.2015) to inform the people about the project and collect the information / suggestions on environmental issues.
- The environmental data was collected within 20 meters of centre of road. The vegetation analysis was done by counting the number of trees within corridor of direct impact (20m from centre of road) and observing the vegetation density along the project road;
- Interaction with other members of the Project Team and APRDC to ensure that environmental considerations were given adequate weight in project planning and design – data and other material from the Inception Report have also been used for the preparation of "Initial Environmental Examination" report.

2 Description of The Project

2.1 Project Type

The State Highway component of proposed VCIC Connectivity Project has considered a length of about 131 km of state highways/MDR which include 4 road sections (Table 2.1).

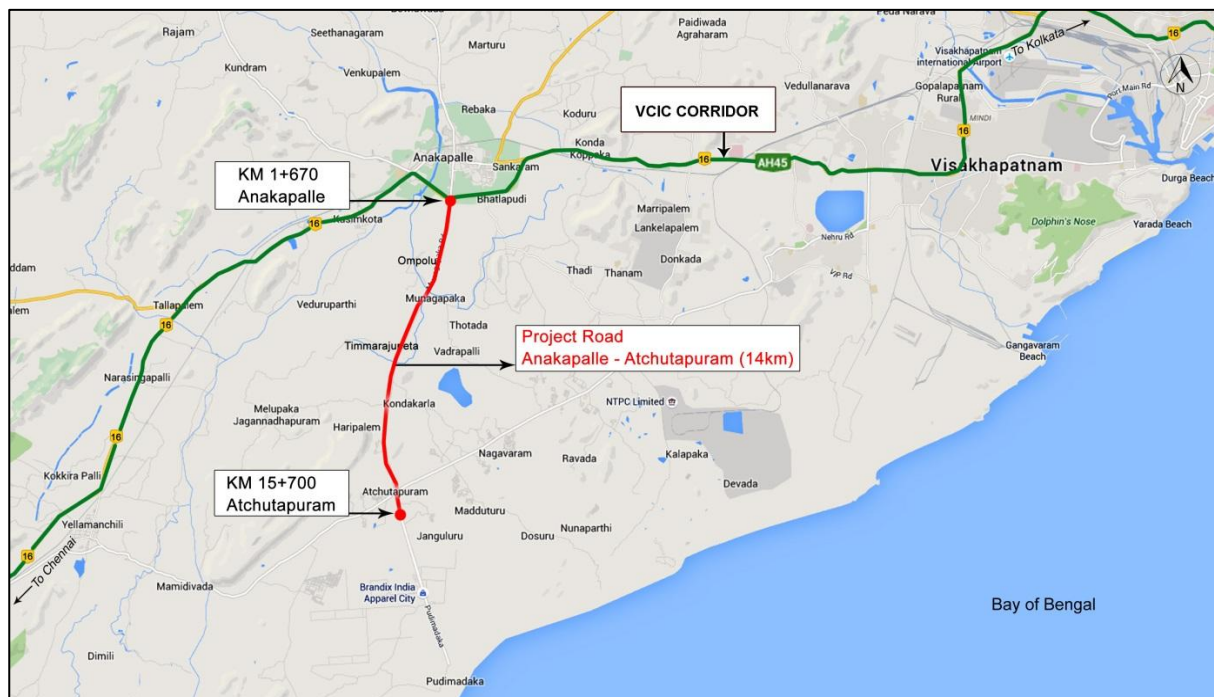
Table 2.1: Roads of VICC Package

As per ToR				
S.NO	Name of the Project Road	District		Length (Km)
1	Anakapalle- Atchutapuram	Visakhapatnam		20
2	ADB Road- Kakinada – Rajanagaram	East Godavari		56
3	Kakinada Anchorage Port - Captive Port in KSEZ - Kakinada Uppada Beach Road	East Godavari		30
4	Krishnapatnam port to Nellore via Muthukur	Nellore		25
Total length in km				131
As per Design				
S.No	Name of the Project Road	From (Km)	To (Km)	Length (Km)
1	Anakapalle-Atchutapuram	1.650	15.150	13.320
2A	Kakinada – Samarlakota	0.000	26.400	26.4
2B	Samarlakota – Rajanagaram	26.400	56.050	29.71
3	Kakinada - Uppada Beach Road	2.350	32.35	30.00
4	Nellore - Krishnapatnam Port Road	2.200	24.845	22.610
Total length in km				122.40

Figure 2—1: Location Map



Map of the VCIC-Phase-1 project roads



Map of Anakapalle to Atchutapuram road

2.2 Need of the Project

The Visakhapatnam-Chennai Industrial Corridor Connectivity Project is vital from the point of connecting various industrial corridors to the National Highways for the ease in transportation of goods and inviting various industries by providing other infrastructure facilities like water, power etc.

Atchutapuram– Anakapalle: This project which connects the cluster to NH5 needs upgrading to meet the needs of existing and upcoming clusters.

ADB Road - Kakinada to Rajanagaram: This project will be an alternate route to Kakinada-Rajahmundry Road and SH 40 to connect Kakinada and Rajahmundry. The project will reduce the travel distance.

Krishnapatnam port to Nellore via Muthukur: This road will provide an alternate route for transit to Krishnapatnam port from Southern regions.

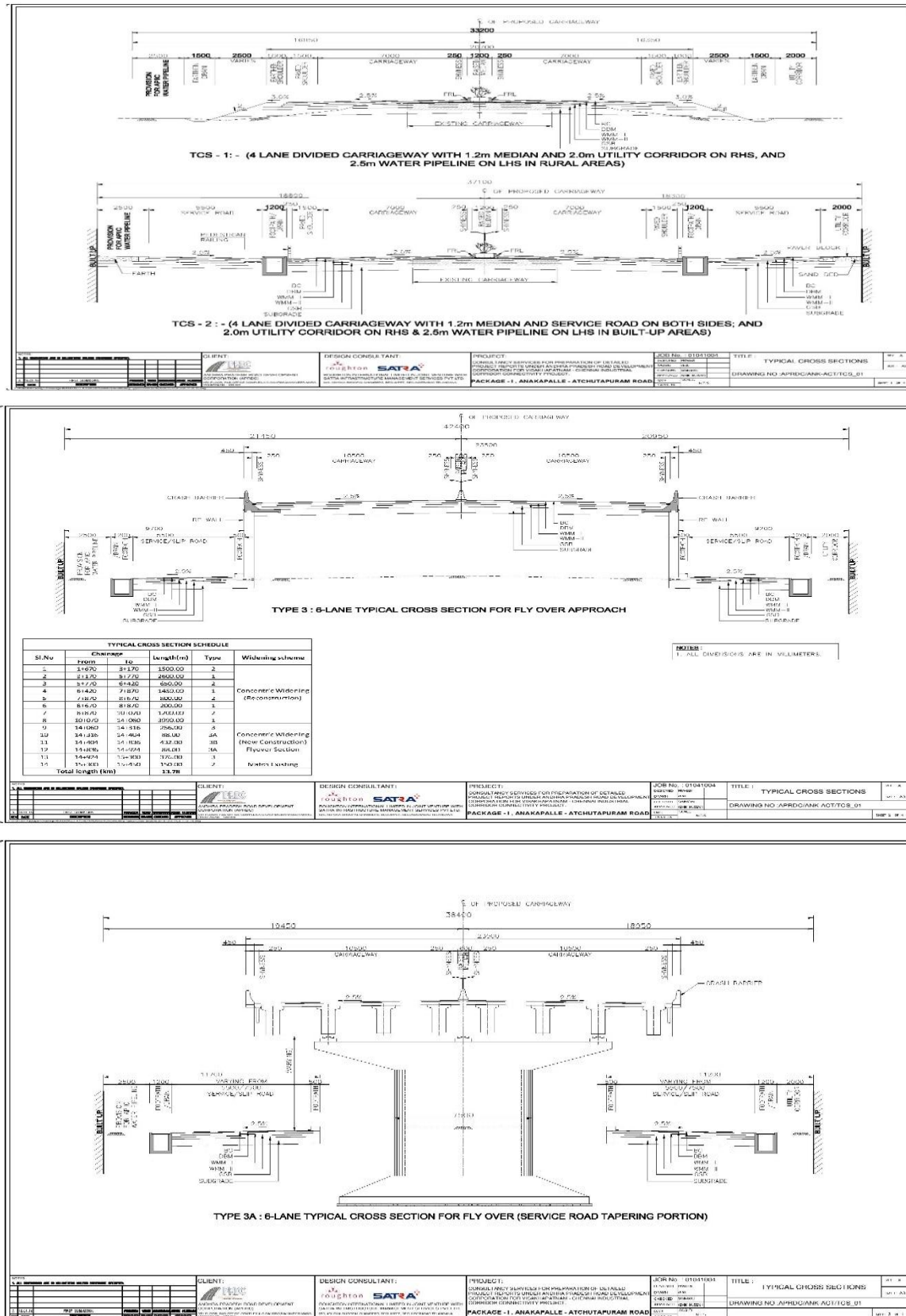
Kakinada Anchorage Port - captive port in KSEZ - Kakinada Uppada Beach Road: The node in Kakinada is close to Uppada, which is connected via Uppada beach road to Kakinada port. Hence, upgrading this road will improve connectivity between nodes to Kakinada port which is the VCIC gateway

The implementation of various project items will have the following direct benefits:

- Improved quality of life for the rural population in the project influence area (10 Km radial distance from the Project road): this as a result of better access to markets, health, education and other facilities; and the derived stimulus for local economic activity.
- A more efficient and safe road transport system through reduced travel times, reduced road accidents, reduced vehicle operating and maintenance costs and reduced transportation costs for goods.
- Intra-state connectivity to Visakhapatnam, Kakinada, Ongole and Nellore districts and also port connectivity to Kakinada port and Krishnapatnam port.

2.3 Sub Project :

The typical cross section of road is shown in figure below:





Before the start of civil works for the any project roads, the APRDC must obtain necessary clearances /permits from the regional office of Ministry of Environment and Forests (MoEF) for Category A project, State Environmental Impact Assessment Authority for Category B project and Consent from State Pollution Control Board. ADB categorises projects based on their potential environmental impacts. This project has been classified as Category B according to the ADB TA Paper. The findings of the Consultant's "Initial Environmental Examination" further confirm that this project falls under **Category B**. This category includes projects with adverse environmental impacts, which are of a lesser degree. As such, an Initial Environmental Examination (IEE) is required and an Environmental Impact Assessment (EIA) may not be required. Category B projects require the preparation of an Environmental Impact Assessment only if "significant" adverse environmental impacts have been foreseen in the IEE. Significant impacts are those, which are comprehensive, broad and diverse and likely to be irreversible. The Rapid Environment Assessment (REA) checklist for this section is appended as *Attachment 1*.

The project road passes mainly through deltaic plain terrain. The topography of the region is mixed type with combination of horizontal and vertical curves. The land use is mixed type with sections of agricultural land on plains and small/medium size industries in clusters. Roadside plantation is observed all along the project road. The density is approx. 55 trees/Km. the improvement work will be undertaken along the existing alignment and the requirement for additional land is being ascertained to accommodate the proposed improvement work. There are no environmentally critical areas in the project road.

As part of IEE, an “Environmental Management Plan” is also prepared. Amongst other things, this plan includes issues associated with “Environmentally Friendly Road Construction”, approach, which is now a need for all the road construction.

3 Description of Existing Environment

A brief description about the existing environment, including its physical and ecological resources, economic development of the region, and issues relating to quality of life are presented in this section. Broad aspects on various environmental parameters (geology, soil, topography, climate, land use, water resources, water quality, air quality, noise quality, tourism, cultural resources etc.) which are likely to be affected (direct or indirect) by the proposed road improvement project are covered. These aspects are covered in broader geographic extent to present the entire project region.

3.1 Physical Environment

Topography, Geology, and Soil

Andhra Pradesh is eighth largest state of the country has a geographical area of 1.6 lakh sq. Km, which constitutes 5.05% of the land area of the country. The project area lies between latitude 17°41'23" North & longitude 83°0'8.45" East and latitude 17°33'50" North & longitude 82°58'44.57" East.

The proposed project road between Anakapalle to Atchutapuram is situated on alluvial coastal plain of south. The soil type in the area is predominantly red loamy soils.

Climate

The climate of Andhra Pradesh varies considerably, depending on the geographical region. Monsoons play a major role in determining the climate of the state. Summers lasts from March to June. In the coastal plain, the summer temperatures are generally higher than the rest of the state, with temperature ranging between 20°C and 41°C.

The month from July to September is the season for tropical rains in Andhra Pradesh. The state receives heavy rainfall from the South-west Monsoon during these months. About one third of the total rainfall in Andhra Pradesh is brought by the North-east Monsoon. The month of October and November see low-pressure systems and tropical cyclones from the Bay of Bengal which along with the Northeast Monsoon, bring rains to the southern and coastal regions of the state. November, December, January, and February are the winter months in Andhra Pradesh. Since the state has a long coastal belt the winters are not very cold. The range of winter temperature is generally 12°C to 30°C.

Visakhapatnam district exhibits a characteristic tropical maritime climate. The average annual rainfall of the district is 1116 mm, and monthly rainfall ranges from nil rainfall in January to 207.5 mm in October. October is the wettest month of the year. The mean seasonal rainfall distribution is 673.5 mm. in southwest monsoon (June-September), 271.8

mm. in northeast monsoon (October-December), 10.9 mm. rainfall in winter (Jan-Feb) and 159.6 mm in summer (March-May). The percentage distribution of rainfall, season-wise, is 60.36% in southwest monsoon, 24.36% in northeast monsoon, 0.97% in winter and 14.3% in summer.

The salient climatic features of the state are as follow:

Average Annual Rainfall	-	1116 mm
Humidity	-	72 to 84%
Wind	-	Light to Moderate
Mean Temperature	-	Summer-32.8-34 ⁰ C Winter-17.5-19.3 ⁰ C

Water Resources and Water Quality

Ground water occurs in almost all geological formations. From the ground water point of view, the aquifers in the district can be broadly classified into hard formations (khondalites, charnockites, granitic gneisses etc.) and soft formations (sand stones and alluvium). Ground water occurs under unconfined to semi-confined conditions in the hard formations, while it occurs under unconfined to confined conditions in soft formations. The yields in the weathered zones of hard formations range from 25 to 100 m³/day. The bore wells drilled in the hard formations, generally tap the fractured and fissured zones. The yields of the bore wells in these formations range between 5 to 25 m³/hr. Sand stones are exposed in the small isolated places around Nakkavanipalem and Elamanchili. In these formations, ground water occurs under both unconfined and confined conditions. The depth of dug wells in alluvium formations ranges from 2 to 10 mbgl and the yields generally ranges from 40 to 250 m³/day. The depth of filter points/tubewells varies from 9 to 35 m with discharges ranging from 15 to 30 m³/hour.

The transmissivity values of the aquifers in the consolidated formations generally vary from 1 to 772 m²/day, whereas specific capacity ranges from 1 to 290 lpm/mdd.

The depth to water level maps show varied water level zones due to underlying terrain and also different geological set up with complex type of hydro-geomorphical structures present in the district. Pre-monsoon (May, 2012) depth to water level map reveals, in general, the water levels are deep particularly in the hilly area of the district. Depth to water levels varies from 5 to 10 mbgl, except at Chintapalli, where water level recorded 15.78 mbgl. In the southern part of the district i.e., near to the coast, the water levels are comparatively shallow (<5.00m) except in Payakaraopeta and Nakkapalli mandals where it is in between 5 and 10m bgl.

The total annual ground water recharge in the district is estimated to be 78,383 ha.m. (Command area = 11,794 ha.m and Non-Command area = 66,689 ha.m.) and the net annual ground water availability in the district after allowing the unavoidable natural discharges is 71689 ha.m (command area 10683 ha.m. and in Non-command area 61,006 ha.m.) The quality of the ground water in the district is, in general, potable and suitable for domestic, industrial and irrigation purposes except at few locations.

Construction of artificial recharge structures like check-dams, contour trenches, percolation tanks and water conservation structures like sub-surface dykes are feasible in the areas where water levels are declining and considerable exploitation of ground water resources is taking place viz. Munagapaka and Payakaraopeta mandals. Rooftop Rainwater Harvesting is to be implemented in the urban areas wherever deepening of water levels are taking place. Along the coast, large number of rain water conservation structures may be constructed to prevent intrusion of seawater. As per the ground water resources of the district all the mandals fall under Safe category, hence no area/mandal has been notified.

(Source: Ground water brochure, Visakhapatnam district, Andhra Pradesh by CGWB, Ministry of Water Resources).

Air Quality

Synonymous with its image of a clean and pristine environment. Nevertheless, location specific issues of air pollution do exist (particularly in the project areas) and air pollution does feature as one of priority environmental issues of the project areas. As the road passes through the agriculture lands and industrial areas, where more number of trees are existing all along the road, the air quality is within the permissible limits.

Dust, vehicular emissions are the main causes of pollution in the road side settlements, villages. Industrial air pollution is mainly confined to areas in the coastal region, where most industries are located. The levels of pollution are far lower and smaller in scale and air pollution is not a significant issue in rural areas because road traffic is very low. In the project area there are no major industrial activities taking place and also the density of traffic on project road is very less. The air quality is reported within permissible limits in these areas.

Negative air quality impacts during construction are likely to result from three main sources, viz. (i) emissions from construction equipment, including delivery trucks; (ii) fugitive dust from earth-moving operations and demolition; and (iii) localised increased traffic congestion in construction areas.

Noise Level

As per the result of reconnaissance survey, it is expected that noise is neither a major issue in the majority of project area nor expected to be a problem except where the project road passes through the settlements, market areas, and junctions at village areas. At busy junctions small contribution to the noise levels are expected, but still the ambient noise quality is expected to be well within the permissible limits.

It is expected that noise will be created during construction activities like operation of heavy machinery, blasting works, the haulage of construction materials to the construction yard and the general activities at the yard itself. Concrete mixing and material movements will be primary noise generating activities and, most likely, will be uniformly distributed over the entire construction period. Noise barriers and other measures - for example the prohibition of certain types of construction activity and the appropriate timing of construction activities - may be required to mitigate these effects.

3.2 Ecological Resources

Vegetation

More than the one third of the area in the District is covered by forest. The forests are of moist and dry deciduous type. The common species available in them are Guggilam, Tangedu, Sirimanu, Kamba, Yagisa, Nallamaddi, Gandra, Vepa etc. Bamboo shrubs are sparsely scattered. But forest area in the district has been showing a quiescent decline since 1955-56 perhaps due to podu practice, indiscriminate grazing and browsing. To stem this, regeneration programmes are being carried out. Chinthapalli Teak Plantation is an off shoot of this. The latest caper in this regeneration programme is rising of Teak, Silver trees, coffee plantations, as the agency areas are found suitable agronomically for coffee growth. Coffee plantations have been raised in about 10,000 Acres in Chinthapalli, Minimuluru, Devarapalli and Ananthagiri regions by different agencies for different purposes. By the forest Department to conserve soil, by the Coffee board to evolve cultures suited to on-traditional areas and by the Girijan Corporation and the I.T.D.A. to wean out tribals from the pernicious practices of "Podu Cultivation."

Andhra Pradesh is endowed with rich and diverse forest resources and it is a reservoir of biodiversity. The forest area of the state is 94689 sq. km constituting 0.71% of the geographical area of the state and 12.44% of the forest area of the country. Legally this area has been classified into "Reserved Forest, Protected Forest and Unclassified Forest", which constitute 65.36%, 32.84% and 1.7% of the forest area respectively.

Details of Trees along the project road

A tree inventory carried out within corridor of impact shows that about 2232 trees (1224 on left side and 1008 on right side) in the direction of Anakapalle to Atchutapuram of various species exist within a corridor of 15 m either side of centerline of road.



Location of Trees along the Road

Wildlife and Protected Area Network

The project road does not pass through the forest area and wildlife/protected area network, and this doesn't exist within Project influence area.

Land Resources

The total geographical area of the district is 11.16 lakh hectares of this 36.45% alone is arable area while 39.53% is forest area. The rest is distributed among "Barren and uncultivable land" about 11.7% and "Land put to non-agricultural uses" about 9.0%.

Agriculture and Forestry

Agriculture is the main stay of nearly 70% of the households. Though Visakhapatnam city is industrially developing, the rural areas continued to be backward. Rice is a staple food of the people and Paddy is therefore the principal food crop of the district followed by Ragi, Bajra and Jowar and Cash Crops such as Sugarcane, Groundnut, Sesamum Niger and Chillies are important. Since there is no Major Irrigation system, only about 36% of the cropped area is irrigated under the Ayacut of the Medium Irrigation System and Minor Irrigation Tanks. The rest of the cultivated area is covered under dry crops depending upon the vagaries of the monsoon. The productivity of the crops is low.

3.3 Economic Development

Fisheries

It is another important economic activity of the fishermen population living in about 59 fishery villages and hamlets on coastline stretching to a length of 132 KMs. covering 11 coastal mandals. About 13,000 fishermen families to take out their livelihood from marine, Inland and brackish water fishing besides catching fish living around Thandava and Raiwada reservoirs.

Transportation

Roadways: The district has a road length of 4721 kms of which the National Highway 5 runs to a length of 134.28 kms. State highway at a length of 355 Kms and balance forms the road maintained by R&B Panchayat Raj Institutions.

Railways: Visakhapatnam Railway Station is a major railway station located in Visakhapatnam, Andhra Pradesh, India. The railway station is under the control of East Coast Railway of Indian Railways. It stands on the Howrah-Chennai main line, though the mainline bypasses Visakhapatnam Central Station. Consequently, a suburban halt on the mainline near Vizag Steel Plant, known as Duvvada, is being developed into a major station to improve services for the city.

Seaways: On the east coast of India, Visakhapatnam serves as the gateway waterway for Andhra Pradesh. It has one of the country's largest ports, and the oldest shipyard on the east coast. It is a land-locked harbour as connected to the sea by a channel cut through rock and sand. Visakhapatnam is one of the busiest ports in India. Vizag Seaport owns two berths in the inner harbour; berth EQ-8 is fully mechanised and berth EQ-9 berth is not. Both berths are capable of handling Panamax vessels. The shipyard at Visakhapatnam is the largest in India. On the Chennai-Kolkata corridor, the city is also a hub for ground traffic. The Gangavaram Port is India's deepest seaport.

Mineral Resources

The District has mineral deposits of Bauxite Apatite (Rock Phosphate) Calcite, Crystalline limestone confined to tribal tracts. Bauxite deposits at Sapparla, Jerrila and Gudem of G.K.Veedhi Mandal are considered to be the largest in the country. Bauxite deposits are also identified at Galikonda, Katuki, Chittemgodndi of Araku group deposits, Katamrajukonda of Gurthedu sub-group of deposits. Phosphate Apatite is available in Kasipatnam village of Ananthagiri mandal. Rich deposits of Crystalline limestone and Calcite are mapped in Borra Caves and along the Valley up to Araku from Borra and around Valasi village of Ananthagiri mandal. Ruby Mica is another mineral available in the District essential for electrical and electronic industries. The mineral occurs in the form of Phologopite and is confined to Borra tract.

Quartz is another mineral found mostly in Bheemunipatnam, Padmanabham, Devarapalli, K.Kotapadu and Ananthagiri mandals. Vermiculate found near Kasipatnam of Ananthagiri mandal. Clay deposits near Malivalasa of Araku mandal are identified. Limeshell useful for manufacture of chemical grade lime is also available in the district. Red and Yellow ochre deposits are also identified in Araku and Ananthagiri mandals

Industrial Situation

Industrial Development is conspicuous in Visakhapatnam urban agglomeration with the large scale industries like Hindustan Shipyard, Hindustan Petroleum Corporation, Coromandal Fertilisers, Bharat Heavy Plates and Vessels, L.G.Polymers Ltd., Hindustan Zinc Plant and the recent giant Visakhapatnam Steel Plant and a host of other ancillary Industries. The Visakhapatnam Steel Plant is the biggest with an authorised share capital of Rs.7466 crores with a licenced capacity of 2.8 Million Tonnes of salable steel 3.0 Million Tonnes of Pig Iron and 8.32 lakhs Tonnes of By product. About 25,000 persons expected to be employed. The project has provided employment to 16300 persons. On the country side the agro based industries like Sugar Factories, Jute Mills and Rice Mills are there besides brick and tile units. There are many Special economic zones and industrial corridors such as, VSEZ, APSEZ,

APIIC, Aganumpudi Industrial Park, Visakha Dairy, JNPC, Brandix Apparel City etc., are located in and around the city.

Tourism and Cultural Resources

Visakhapatnam is famous for its natural harbor, ship building unit, Steel plant, Oil refineries. Dolphins Nose hills, Rama Krishna beach and Rushi Konda Beach and more over it is a popular and famous industrial city in Andhra Pradesh. Bhimunipatnam is 40 km from the Visakhapatnam and is famous for natural sea port. Simhachalam lies in outskirts of the main city. It is famous for the srivaraha Narasimha swamy temple. Araku valley is 112 km away from Visakhapatnam. It is famous for the scenic beauty it offers to the travelers by Train or Road. The area of the valley is roughly 36 km², and the altitude is between 600 and 900 meters above sea level. It is renowned for its association with the Tribal culture & customs. Thotlakonda lies to the north of Visakhapatnam which consists of a Buddhist Monastic Complex remnants and rock cut cisterns on the hilltop at mangamaripeta. Bavikonda is another famous Hilltop Buddhist Monastic Complex remnant site near to Thotlakonda along the beach road to Bhimili from Vizag. Gudilova is famous for the temple of Lord shiva and Lord Ranganatha swamy located on a small hillock between two big hills in outskirts of Visakha city along the Eastern Ghats, it is a natural picnic spot also. Located on the state highway of Pendurthi and 7 km from Anandapuram of Visakhapatnam.

The Borra Caves, also called Borra Guhalu. In Telugu language "Borra" means something that has bored into the ground and "guhalu" means (caves) are located on the East Coast of India, in the Ananthagiri hills of the Araku valley. The Kondakarla Ava, One of the largest freshwater lake in the country and cradled in the foothills of the Eastern Ghats, the lake Konda Karla-Ava is situated at 50 Kilometer from Vizag on sankaram (Buddhist Place) - Etikoppaka (Craft Village) - Kondakarla Ava Circuit. Kondakarla Ava is the second largest fresh water lake in the state. It has been recognized as Eco Tourism destination. Kondakarla Ava is home to an array of flora and fauna.

Energy and Electric Power Potential

There are several power plants in and around Visakhapatnam district. Simhadri Super Thermal Power Plant of NTPC Limited is expanding from 1,000 to 2,000 MW at a cost of ₹ 50 billion (US\$740 million). Hindujas has begun construction of a 1,070-MW thermal power plant in Visakhapatnam district at a cost of ₹70 billion (US\$1 billion).

NTPC is establishing 4x1,000 MW imported coal-based thermal power plant in Visakhapatnam district in Andhra Pradesh, which will come up at an investment of Rs 20,000 crore. Approximately Rs 5 crore outlays are needed for generation of one megawatt thermal power.

3.4 Quantum of construction materials required and Borrow area particulars

The different materials required for constructions of proposed road are as mention below table

Table 3-1: Quantum of construction materials required

Anakapalle - Atchutapuram Road (From Km 1+670 to Km 15+450)			
S. No	Item Description	Unit	Approx. Quantity
1	Fine Aggregates	Cum	22,461
2	Course Aggregates	Cum	322,709
3	Steel	MT	4,800
4	Cement	MT	19,625
5	Bitumen	MT	6,070
6	Barrow Quantity	Cum	6,75,000

The quantum of borrow soil required is 6,75,000 cubic meters of soil. The borrow areas; locations have been identified by the consultants. The below Table 3-2 clearly informs that 3 borrow area locations identified for this sub project, in which 3 locations are Government land identified. The area of land identified for borrow soil is about more than 6 Acres. All the borrow areas are nearer to the proposed road in the vicinity of less than 2 KMs only and easy for the transportation.

All the locations are isolated and do not have any problem to the habitations, forest areas. The locations photos are being furnished with this report. The detailed description has been explained in the quantities requirement chapter in the DPR report.

The quarries are available in the vicinity of proposed road, to get the fine aggregates, course aggregates etc., they are also located in isolated areas and does not have any negative impacts either to the habitations or to the environment. All the prescribed precautions will be taken care by the authorities concerned during the transportation period of work execution.

Table 3-2: Details of Borrow Areas

B. A. No.	Chinage (km)	Lead (km)	Side	Ownership Details	Available Quantity
			(LHS/RHS)		
B.A-1	7+100	0.4	LHS	Govt. Land.,	Hills - Adequate
				Chuchukonda Village	
B.A-2	9+000	2.1	RHS	Govt. Land.,	Hills - Adequate
				Vadrapalle Village	
B.A-3	11+700	1.3	LHS	Govt. Land.,	Hills - Adequate
				Chowdapalle Village	

Figure 3—2: Geological & Mineral Map of Andhra Pradesh

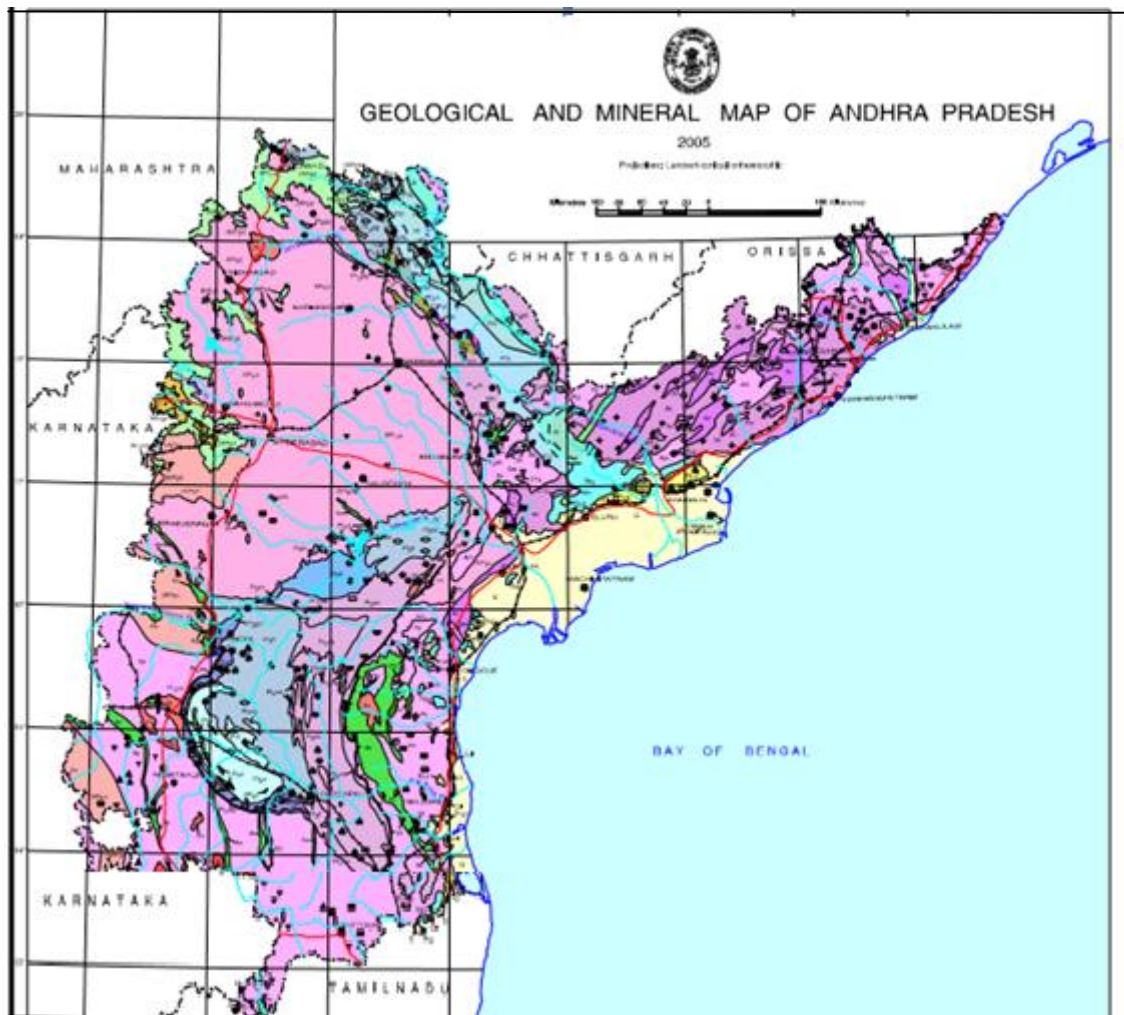
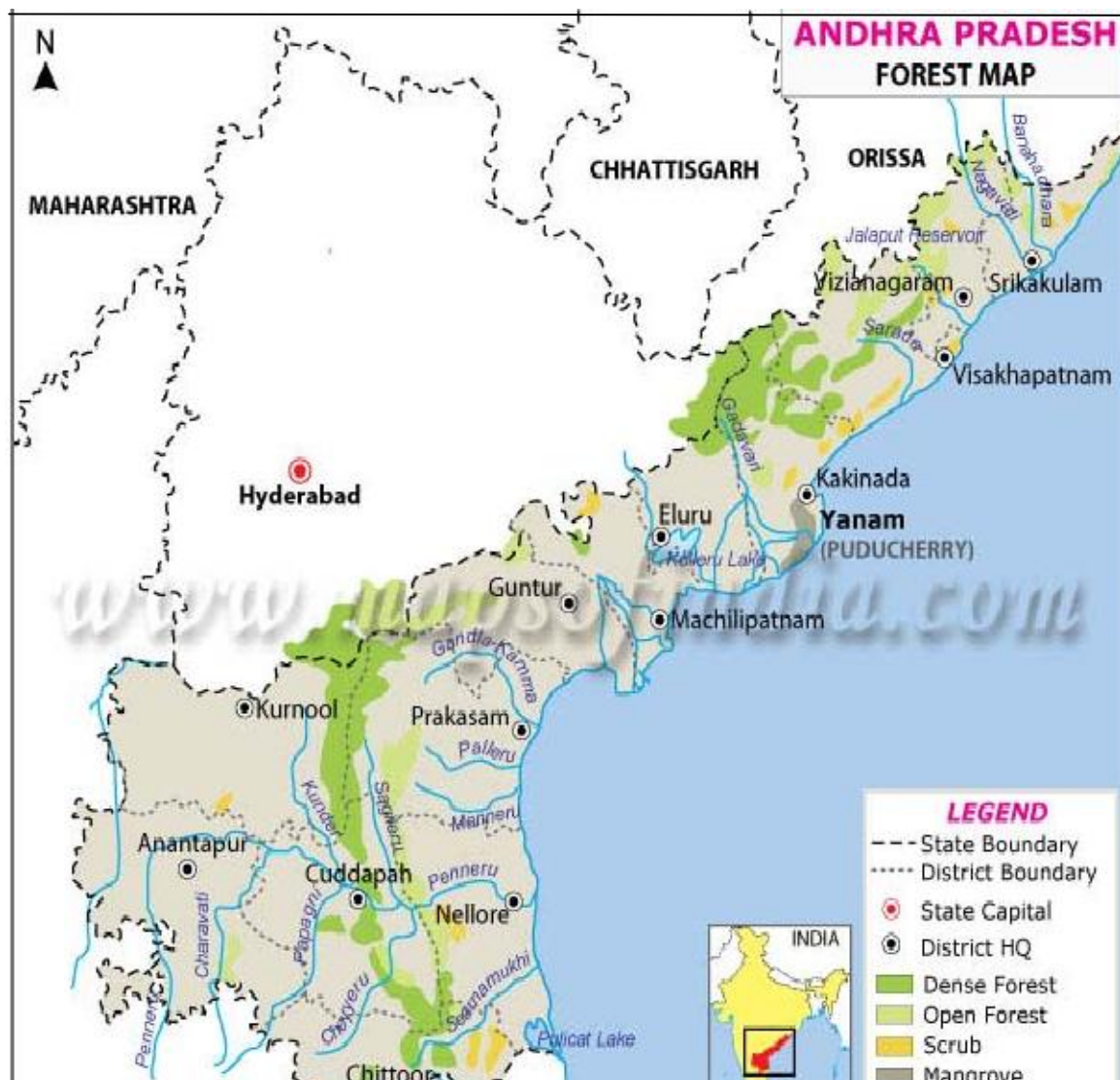


Figure 3—3: Forest Map of Andhra Pradesh



4 Policy, Legal and Administrative Framework for Environmental Management in India

4.1 Policy

As a sequel to the UN Conference on the Human Environment (1972), Indian parliament in 1976 amended the Constitution of India by introducing articles 48A and 51A. These articles incorporated environmental concerns into the Directive Principles of state policy and postulated as a fundamental duty of all citizens to preserve and protect the environment. As per the constitutional provisions:

The state shall endeavour to protect and to improve the environment and safeguard forests and wildlife of the country (Article 48A) and

Every citizen is bound to protect and improve the natural environment and to have compassion for living creatures [Articles 51A (g)].

4.2 Legislation

Government of India has enacted more than thirty environment conservation laws and Acts. Some of these are: The Environment (Protection) Act 1986, The Wildlife (Protection) Act 1972; The Forest (Conservation) Act 1980; The Water (Prevention and Control of Pollution) Act 1974, The Air (Prevention and Control of Pollution) Act 1981; 1988 amendment of The Motor Vehicle Act (M.V.) Act, 1939. The Water and The Air Acts entrusted the task of their implementation and regulation to pollution control boards set up for such purpose at State and Central levels.

Even prior to the 1970's and 80's flurry of environmental legislation, there existed numerous statutory provisions for safeguarding environment, though in a highly limited way as some of these dated back to the colonial period. Some of these were Section 277 of IPC dealing with water pollution, Section 278 of IPC dealing with atmospheric pollution and Statutes like the Factories Act, 1948; the Mines Act, 1952, Insecticides Act, 1968 and Motor Vehicle Act (M.V.) 1939.

Some of the important acts applicable to the present project are described herewith:

The Environment (Protection) Act, 1986

The Environment (Protection) Act, 1986 is widely regarded as a comprehensive or umbrella legislation for environment in its entirety. The responsibility for implementation of the provisions of the EPA has to a large extent been entrusted to the regulatory agencies created under the Air and Water Acts. Department of Environment (DoE) was created in 1981 in the Central Government to act as a nodal agency for environmental protection and development in a co-ordinated manner.

The principal environmental regulatory agency in India is the Ministry of Environment and Forests (MoEF) of the Government of India. MoEF formulates environmental policies and accords environment clearances for the large projects (sector-wise listing done by MoEF). In the North Eastern Region, the State Department of Environment and State Pollution Control Boards are enforcing authorities at the State level to resolve environmental issues arising due to any infrastructure project. MoEF has a regional office at Shillong to look after the matters related to environment in the region.

The Environmental Impact Assessment Notification (2006) and Amendments

The Environmental (Protection) Act, 1986 provided for Environment (Protection) Rules, which are formulated since then. As one of these rules, the Environmental Impact Assessment Notification 2006 has identified highways (item 7f of Schedule) as one of the project requiring prior clearance from the central government / state government. Environmental Impact Assessment (EIA) is a statutory requirement for obtaining clearance (a comprehensive format for EIA has been prescribed in the notification).

According to the notification from the MoEF dated 14th September 2006 and subsequent amendments, i) New State Highways; and ii) Expansion of National / State Highways greater than 30 km involving additional right of way greater than 20m involving land acquisition; are classified as Category B projects. Hence, project requires prior environmental clearance from State Environmental Impact Assessment Authority (SEIAA). An application seeking prior environmental clearance in all cases shall be made in the prescribed Form 1 and supplementary Form 1A, if applicable; annexed to the said notifications.

Forest (Conservation) Act as amended in 1980

Of all laws, the Forest (Conservation) Act, 1980 stands out as having particular significance for this project. The Act pertains to the cases of diversion of forest land and felling of roadside plantation. Depending on the size of the tract to be cleared, clearances are applied for at the following governmental levels.

Applicability of the Forest (Conservation) Act to the Roadside Strip Plantations

In 1986, when Ministry of Environment & Forests enacted the Environment Protection Act, the entire linear stretches of roadside plantations along the State Highways were declared as protected forest. Although the land is under the control of the PWD, due to its protected status, clearance is required to cut roadside trees. Applicability of the provisions of the Forest (Conservation) Act, 1980 to the linear (road or canal side) plantations was modified by a notification from the MoEF,

dated 18 February 1998. The new notification recognized that the spirit behind the Forest (Conservation) Act was conservation of natural forests, and not the strip plantations. In the case of the "notified to be protected" roadside plantations, the clearance now may be given by the concerned Regional Offices of the MoEF, irrespective of the area of plantation lost. While issuing the approval, in place of normal provision for compensatory afforestation, the Regional Offices will stipulate a condition that for every tree cut at least two trees should be planted. If the concerned Regional Office does not accord the clearance within 30 days of the receipt of fully completed application, the proponent agency may proceed with the widening/expansion under intimation to the State Forest Department, and the MoEF, Government of India.

Forest Land

Restrictions and clearance procedures proposed in the Forest (Conservation) Act apply wholly to the natural forest areas; even in case of the protected/designated forest area does not have any vegetation cover.

The proposed Project road does not involve any acquisition of forest land, hence clearance from Central Government is not required.

The Wildlife (Protection) Act 1972

This act has no relevance to the proposed road project, as there are no wild-life protected areas/sanctuaries/national parks in the project influence area.

The Water (Prevention and Control of Pollution) Act 1974 (Amended 1988)

The Water (Prevention and Control of Pollution) Act, 1974 resulted in the establishment of the Central and State level Pollution Control Boards whose responsibilities include managing water quality and effluent standards, monitoring water quality, prosecuting offenders and issuing licenses for construction and operation of certain facilities.

The Air (Prevention and Control of Pollution) Act 1981 (As Amended in 1987)

In the state, Andhra Pradesh Pollution Control Board is empowered to set air quality standards and monitor and prosecute offenders under the Air (Prevention and Control and pollution) Act, 1981.

The responsibility for monitoring vehicular - air and noise pollution lies with the State Transport Authority (STA), not with the APPCB.

The Government of India (GoI) standards for the discharge of pollutants to the environment (Water and Noise standards) are listed under the Environmental Protection Act. These standards apply mainly to the control of industrial pollution, with some apply to road construction projects.

Before implementation of the project, it is mandatory requirement to get NOC from Andhra Pradesh Pollution Control Board.

In order to obtain an environmental clearance from the MoEF, "Public Hearing" was previously mandatory, as per the MoEF notification, which came into effect on 10th April, 1997. However, revised notification of the MoEF, exempts highway widening / up-gradation projects from public hearing.

The Noise Pollution (Regulation and Control) Rules, 2000

The Union Government has laid down statutory norms to regulate and control noise levels to prevent their adverse effects on human health and the psychological well-being of the people. The rules titled Noise Pollution (Regulation and Control) Rules, 2000 have come into force at February 14, 2000. Under the new regulation, different areas and zones are to be identified as industrial, commercial, and residential or silence areas and anyone exceeding the specified noise level would be liable for action. In industrial areas, the noise level limit during the day time (6 am to 10 pm) is 75 decibels and during night (10 pm to 6 am) is 70 decibels.

Similarly, for commercial areas day time limit is 65 decibels and night limit is 55 decibels. In the case of residential areas, the limits are respectively 55 and 45 decibels and for the silence zones, 50 and 40 decibels.

The Motor Vehicles Act 1988

In 1988, amendment of the Indian Motor Vehicle Act empowered the State Transport Authority to enforce standards for vehicular pollution prevention and control. The authority also checks emission standards of registered vehicles, collects road taxes, and issues licenses.

In August 1997, the "Pollution Under Control" (PUC) programme was launched in an attempt to crackdown on the amount of vehicular emissions in the state. To date, it has not been highly effective.

The Movement of Hazardous Chemicals

Movement of hazardous chemicals by road is governed by Central Motor Vehicle Rules, 1989 (rules 129 through 137). Besides, regulations and precautions have to be taken while transporting such goods, the rules stipulate availability of a Transport Emergency (TREM) Card with the driver of the carrier which shall provide information on hazardous nature of the chemical carried and also precautions required to handle emergencies such as spillage and fire.

4.3 Environmental Administrative Framework

The MoEF has the overall responsibility to set policy and standards for the protection of environment along with the Central Pollution Control Board (CPCB). This includes air, noise and water quality standards and the

requirements for the preparation of Environmental Impact Assessment (EIA) statements for development projects. These standards are of significance for the proposed project. The status of key environmental legislation in India is given in Table 4-3. Salient features of relevant environmental laws and regulations, including their applicability to this project is given below.

Table 4-3: Statutes, Policies, Regulations and Responsible Agencies

Sl. No	Agency	Statute/Policy	Relevant objectives
1	Ministry of Environment and Forests	Environment (Protection) Act 1986	To protect and improve the quality of the environment and to prevent, control and abate environmental pollution
		Forest (Conservation) Act, 1927 Forest (Conservation) Act, 1980 (as amended in 1998) Forest (Conservation) Rules, 1981	To restrict deforestation by restricting clearing of forested areas
		Environmental Impact Assessment Notification 2006 Environment Protection Rules, 1986	To ensure that appropriate measures are taken to conserve and protect the environment before commencement of operations
2	Pollution Control Boards (State)	Water (Prevention and Control of Pollution) Act 1974 as amended in 1988 Air (Prevention and Control of Pollution) Act 1981 as amended in 1987	To provide for the prevention and control of water pollution and the maintaining or restoring wholesomeness of water To provide for the prevention, control and abatement of air pollution and for the establishment of Boards to carry out these purposes.
3	Environment and Forest Department	Wildlife (Protection) Act, 1972	To protect wild animals and birds through the creation of National Parks and Sanctuaries
4	Department of Transport	Motor Vehicle Rules, 1989 Motor Vehicles Act, 1988	To check control vehicular air and noise pollution

Sl. No	Agency	Statute/Policy	Relevant objectives
	and Department of Police	Rules of Road Regulations, 1989	To regulate development of the transport sector
5	Archaeological Survey of India, Directorate of Archaeology	Ancient Monuments and Archaeological sites and Remains Act, 1958	To protect and conserve cultural and historical remains To regulate construction activities near the monuments and sites protected by the Government
6	Revenue Department	Land Acquisition Act, 1894	To set out rules for acquisition of land by the Government departments and agencies

Other guidelines available in respect of highway projects are:

- Indian Road Congress IRC 104 : 1998
- ADB's Environmental Assessment Guidelines (2003)
- ADB's Guidelines on Environmental Assessment Guidelines of selected infrastructure projects (2000)
- ADB's Environment Policy, the Operations Manual F1/BP and F1/OP (2003)
- ADB's Safeguard Policy Statement (2009)
- Ministry of Road Transport and Highways (MORTH)

4.4 National Environmental Quality Standards

Air Quality Standards

In order to evaluate air quality and to design appropriate air pollution control systems, it is necessary to know the concentration of various air pollutants. The guidelines issued by CPCB on ambient air quality standards are reproduced in the Table 4.4. It needs to be mentioned here that for HC, ambient air quality standards have not been specified.

Table 4-4: National Ambient Air Quality Standards

Pollutant	Time weighted average	Sensitive area	Industrial area	Residential, rural & other areas	Method of measurement
Sulphur Dioxide (SO ₂)	Annual* 24 hours**	15 µg/m ³ 30 µg/m ³	80 µg/m ³ 120 µg/m ³	60 µg/m ³ 80 µg/m ³	Improved West and Gaeke Method Ultraviolet Fluorescence
Oxides of Nitrogen as NO _x	Annual* 24 hours**	15 µg/m ³ 30 µg/m ³	80 µg/m ³ 120 µg/m ³	60 µg/m ³ 80 µg/m ³	Jacob & Hochheiser Modified (Na-Arsenite) method Gas phase Chemiluminescence
Suspended Particulate Matter (SPM)	Annual* 24 hours**	70 µg/m ³ 100 µg/m ³	360 µg/m ³ 500 µg/m ³	140 µg/m ³ 200 µg/m ³	High Volume Sampler (Average flow rate not less than 1.1 m ³ /minute)
Restorable Particulate Matter (RPM) size less than 10 µm	Annual* 24 hours**	50 µg/m ³ 75 µg/m ³	120 µg/m ³ 150 µg/m ³	60 µg/m ³ 100 µg/m ³	Respirable Particulate Matter Sampler
Lead (Pb)	Annual* 24 hours**	0.5 µg/m ³ 0.75 µg/m ³	1.0 µg/m ³ 1.5 µg/m ³	0.75 µg/m ³ 1.0 µg/m ³	AAS Method after sampling using EPM 2000 or equivalent filter paper
Carbon Monoxide (CO)	8 hours** 1 hour	1.0 mg/m ³ 2.0 mg/m ³	5.0 mg/m ³ 10 mg/m ³	2.0 mg/m ³ 4.0 mg/m ³	Non - dispersive infrared Spectroscopy

Source: Central Pollution Control Board, 1997

Annual arithmetic mean of minimum 104 measurements in a year taken twice a week 24 hourly at uniform interval. 24 hourly/8 hourly values should be met 98% of the time in a year. However, 2% of the time, it may exceed but not on two consecutive days

Water Quality Standards

The project that crosses waterways viz. rivers, canals, streams, etc can have significant impacts on both surface and groundwater hydrology. A change in water hydrology may affect the surface water quality as well as sediment transport, changes in water table, water logging and changes in infiltration rates. The excerpts from guidelines issued by CPCB (based on

BIS standards) on primary water quality have been reproduced in Table 4.5.

Table 4-5: Guidelines of CPCB on Primary Water Quality

Designated Best Use	Class of Water	Criteria
Drinking water source (with conventional treatment)	A	Total Coliforms MPN/100ml shall be 5000 or less pH between 6 to 9 Dissolved Oxygen 4 mg/l or more Biochemical Oxygen Demand (BOD) 5 days 20°C 3 mg/l or less
Outdoor bathing (organised)	B	Total Coliforms MPN/100ml shall be 500 or less pH between 6.5 to 8.5 Dissolved Oxygen 5 mg/l or more Biochemical Oxygen Demand (BOD) 5 days 20°C 3 mg/l or less
Drinking Water Source (without conventional treatment)	C	Total Coliforms MPN/100 ml shall be 50 or less pH between 6.5 to 8.5 Dissolved Oxygen 6 mg/l or more Biochemical Oxygen Demand (BOD) 5 days 20°C 2 mg/l or less
Propagation of Wildlife	D	pH between 6.5 to 8.5 for Fisheries Dissolved Oxygen 4 mg/l or more Free Ammonia (as N) 1.2 mg/l or less
Irrigation, Industrial Cooling, Controlled Waste	E	pH between 6.0 to 8.5 Electrical Conductivity at 25°C Max 2250µ mhos/cm Sodium absorption ratio Max. 26 Boron, Max. 2 mg/l

Noise Standards

The MoEF has notified ambient noise level standards vide Gazette Notification dated 26th December 1989. It is based on the weighted equivalent noise level (Leq). These are presented in Table 4.6.

Table 4-6: National Ambient Noise Standards

Area Code	Category of Zones	Limits of Leq in dB(A)	
		Day time*	Night time*
A	Industrial	75	70
B	Commercial	65	55
C	Residential	55	45
D	Silence Zone **	50	40

* Day time is from 6 am to 9 pm whereas night time is from 9 pm to 6 am

** Silence zone is defined as area up to 100 meters around premises of hospitals, educational institutions and courts. Use of vehicles horns, loud speakers and bursting of cracking are banned in these zones

As mentioned in section 3.1.5, these noise standards have been given the status of statutory norms vide Noise Pollution (Regulation and Control) Rules, 2000. However, these rules have changed the periods for 'Day Time' and 'Night Time' to 6 a.m. to 10 p.m. and 10 p.m. to 6 am respectively.

5 Screening of Potential Environmental Impacts and Mitigation Measures

5.1 Introduction

This section presents the key environmental issues associated with different aspects of the proposed project. Identification and assessment of the potential environmental impacts are based on secondary information supplemented by data collected by field visits. Impacts on various environmental components have been assessed at four different stages namely Project location, Design, Construction and Operational stages.

The implementation of a road project can cause environmental impacts that are short, or long-term, and beneficial, or adverse, in nature. As it is further discussed, the overall long-term impacts will be largely beneficial. The proposed Anakapalle to Atchutapuram road section will have major positive impacts in regard to the socio-economic environment and quality-of-life particularly in Anakapalle and Atchutapuram towns of Visakhapatnam district.

A few short and long-term negative effects, mainly at the construction and operation stages, are, nonetheless, anticipated. These can, however, be kept in check through proper planning and adopting Environment Friendly Road Construction methods and the appropriate regulatory measures.

The present section on identification and assessment of impacts has been structured in sequence to Chapter 3 "Description of the Environment". The project activities are generally described first and the impacts are discussed thereafter.

5.2 Identification and Assessment of Impacts

Positive Environmental Impacts due to improvement of highway sections between Anakapalle and Atchutapuram are:

- Improvement of the Anakapalle- Atchutapuram section to Four-lane configuration will result in connectivity to the National Highway, reduction in travel time and lower vehicle operating cost i.e. per kilometre vehicle operating cost from the general improvement work and an absolute saving in cost due to reduction in fuel consumption for the existing traffic.
- Smooth flow of traffic can be anticipated once the four lane road is laid.

Negative Environmental Impacts due to improvement of highway sections between Anakapalle and Atchutapuram are:

- Improvement of project road may require cutting of road side trees that falls within formation width i.e. 15.0m. The cutting may reduce

the ecological balance of the area and also enhance soil erosion problem. Also, run-off from the road and the re-channelization of existing drainage may increase erosion. The noise, air and water pollution and disposal of waste, during construction, may negatively impact both local residents and domestic animals. These latter effects should, however, only be temporary/reversible.

- There will be a number of quarries and other sources as suitable sources for construction materials. The operation of these quarries, transport of material from quarry to construction site and the scars the quarries may leave on the landscape once work is complete are, however, matters that must be carefully addressed if negative impacts are to be minimised. However, the operation of quarries is an independent and already regulated activity by the Department of Mines & Geology.
- Also Improvements on road and construction of bridges may enhance soil erosion, and reduce the micro-level ecological balance of the area. Construction will also disturb the habitation of fauna living in this area. These should, however, be only temporary/reversible effects. The widening will also require clearing of vegetative ground and cutting of trees (tentatively 4251 nos.).

The likely negative impacts of various aspects of project on the surrounding environment are more fully described in the following sections.

5.3 Negative Impacts :

Related to Project Location, Preliminary Planning and Design

Location issues

There will be a requirement to establish construction camps and related contractor's facilities, borrow pits and quarries. These must be located in environmentally sound and socially safe areas. It is expected that construction materials for the road works will be mined only from approved quarries. The following criteria must, however, be applied when locating borrow areas:

- Borrow areas are not to be established in ecologically sensitive areas;
- Villagers are to be consulted with respect to location of all borrow areas – these should ensure the safety of local communities and, if possible, should incorporate beneficial post construction features for the villages;
- Borrow areas are to be located away from the Corridor of Impact of Project road as well as 500m away from settlements, so as to minimise visual impacts.

In regards to the setting-up of construction camps for labourers:

- These should be located at least 500m away from settlements;
- Living accommodation and ancillary facilities should be erected and maintained to standards and scales approved by the Engineer-in-Charge;
- Toilets and urinals should be provided in accessible places away from the Hot mix plant and mixing yard; and

There are no adverse impacts expected on historical places/monuments. However, there are few religious structures/idols which are coming within ROW and adjacent to existing carriageway. Care must be taken to relocate these structures. Also earthworks, associated with the actual road construction/improvement works, or deriving from secondary sites such as quarries or borrow pits, may reveal sites or artefacts of cultural/archaeological significance. In the event of such discovery, the concern authorities (Archaeological Survey of India) should be informed and the requirement to take such action should be incorporated in contract documents.

The project road does not pass through (or) adjoining to the protected area (or) forest land (or) ecologically sensitive area. To minimize loss of vegetative cover and/or trees and soil erosion the following mitigation measures are adopted during the detailed design and construction stage of the project:

- The detail engineering design study shall be carried out for widening proposal in order to minimize the need for tree felling or removal of vegetation.
- The detailed engineering design shall identify areas prone to erosion and include land stabilization as part of the design.
- The detail engineering studies and construction activities will strictly enforce the environmental conditions put as part of the Environmental clearance and Consent conditions from the SPCB.
- The improvement of road will be done by adopting Environmental Friendly Road Construction (EFRC) methods.

The improvements to the existing road for 4-lane configuration will require the need to cut vegetation along the project road. This will inevitably have a more significant impact and this matter is discussed in the following sections.

Other Impacts deriving from the Project Preliminary Planning and Design Process

During preliminary planning and design of this project, the Consultants have taken into account the need for:

Optimum siting and control of quarries; providing adequate cross-drainage structures/drains;

Providing side-drainage structures; Mechanised construction methods and thereby, for example, reduced use of firewood for heating bitumen; Maximising safety and thereby reducing traffic accidents; Reducing travel times and, thereby, fuel consumption and emissions; Increased accessibility for residents to education, health facilities, markets etc., and for others who might come as tourist or other purposes; and Improving the socio-economic conditions of residents in the project area of influence.

As part of the engineering works for this work, the following have been the guiding principles in determining preliminary alignments and other matters concerning route. These principles are implemented during detailed design wherever possible and, if this is not possible, the appropriate adjustments should be made on site during the construction phase:

S.No.	Environmental Issue	Measures to be taken
1	Alignment	Final alignment should be determined so as to minimise land acquisition and the impact on people, animals and to avoid unfavourable geological condition and cultural relics.
2	Soil erosion	Temporary and permanent drainage systems should be designed to minimise the soil erosion.
3	Dust and air pollution	Borrow sites, waste disposal sites and asphalt mixing sites should be identified – keeping in mind environmental issues such as dust generation & noise pollution.
4	Cultural heritage	Any archaeological sites/remains identified along the alignment should be intimated to ASI prior to construction.

5.4 Construction : Permits and Environmental Impacts to be taken into Particular Account during Construction

Permits

As a mandatory requirement of Environmental Impact Assessment Notification, September 2006, by Government of India, any development activities should not be taken in any part of country unless it has granted clearance from concerned authority.

Highways are classified as one of the project (Item 7f), listed in said notification, which require prior clearance. The proposed project is classified under 'category B' project as per the said notification and need prior environmental clearances from the State Level Environmental Impact Assessment Authority (SEIAA) at State Level.

At present, No clearances for the Project road has been obtained so far. The process is to ensure minimal negative environmental effects. Prior to applying for environmental clearance, a number of resource-specific clearances and permits have to be obtained. The following Table 5-7 outlines the type of clearances and permits and the authorised bodies that issue them along with the procedures involved.

Table 5-7: Type of clearances and permits

S.No.	Clearance/ permit	Authorised body	Procedures involved	Time involved
1.	Environmental Clearance	State Level Environmental Impact Assessment Authority, Government of Andhra Pradesh	Detailed Proposal in specified format given in Appendix 1 - Form I of EIA notification 2006.	About 6 months
2.	NOC from Andhra Pradesh Pollution Control Board	Member Secretary, Andhra Pradesh Pollution Control Board	Application in prescribed form along with project report and required fee shall be submitted to APPCB. After scrutiny of application if required, concerned Pollution Control Board shall conduct public hearing in particular district involving State authorities and Forest department and affected persons. After fulfilling the requirement SPCB may issue the NOC with specific conditions.	Between 6 month to 1 year
3.	Clearance for quarry sites	Department of Geology and Mines, Govt. of Andhra Pradesh	Submission of application for stone quarry to the Department of Mines and Geology after scrutiny of application and consultation with revenue department together with site verifications will give approval with specific conditions.	Takes between 3-6 months.

Unlike other states, in Andhra Pradesh, the road side plantations within right of way, comes under the authority of R&B Department. But for felling & transportation of trees, permission has to be taken from the Forest department at DFO level, under Andhra Pradesh Water, Land & Trees Act (WALTA), 2002.

Andhra Pradesh Water, Land Trees Act, (WALTA) 2002: The Act came into force on April 19, 2002 with an objective "to promote water conservation, and tree cover and regulate the exploitation and use of ground and surface water for protection and conservation of water sources, land and environment and matters, connected therewith or incidental thereto".

5.5 Environmental Impacts : To be Taken into Account During Construction

General

Soil and waste generated at construction camps will need to be properly disposed, otherwise it may create odour, other nuisances and disturb local drainage systems.

Quarries and borrow pit sites and the need for such have not been finalised. Localised problems may, however, be expected at the finally selected sites. These problems may include dust and noise pollution and respiratory problems to both workers and local residents.

The improvements to the road section may involve the cutting of some hill slopes and the destruction of some trees. At few location amount of cut and fill work expected to be significant mainly at curves and bridge locations. It also involves uprooting of trees. Debris generated during hill slope cutting should be disposed off in proper places.

The likely environmental impacts from construction activities are summarised in the following sections.

Physical Environment

I. Topography, Geology and Soil

During the improvement works of the road sections and because of felling of trees, ground clearing, stone quarrying, and construction of structures etc. the micro-level topography will change. With proper planning, these topographical impacts can be kept within acceptable limits and sometimes even used to enhance local aesthetics. Any negative impacts on topography (existing or new), particularly soil erosion due to a lack of drainage facilities, will be minimised with the provision of proper drainage facilities such as culverts, causeways etc. The overall impact on topography is, therefore, anticipated to be insignificant.

During construction phase, following restrictions should be imposed:

- Existing vegetation including shrubs and grasses along the road (except within the strip directly under embankments or cuttings) should be properly maintained;
- Sites for quarrying, borrowing and disposal of spoils are to be confirmed according to the applicable laws and regulations in the state and the practices followed in recent/ongoing internationally funded road projects should be continued;
- Controlled and environmentally friendly quarrying techniques should be applied to minimise erosions and landslides;
- Blasting (if any) should not be carried out during busy periods;
- Cut material should be disposed of in suitable depressions;

It is also important to:

- Maintain adequate vegetative cover above and below the road;
- Maintain the natural course of water bodies (that is as far as possible) and avoid throwing debris into stream courses;
- Construct proper drainage structures, in order to prevent soil from being saturated with water and hence susceptible to erosion;

II. Erosion, Silt Run-Off and Landslides

Excavation and earthworks should be mainly undertaken during the dry season when the risks from erosion and silt run-off are least. The materials used for surface dressing will consist of aggregates and gravel, which do not contain silt. Internationally accepted best practice engineering approaches to minimise landslide and erosion risks and silt run-off will be incorporated into contract documents and monitored during construction.

In order to minimise erosion, and silt run off, it will also be important to:

- ensure that all embankment grades are not too steep and prone to erosion;
- waste material is not thrown into nearby river streams and cross cutting water bodies;
- temporary retention ponds, interception drains, and silt traps are installed to prevent silt laden water from entering adjacent water bodies;
- the topsoil of borrow areas is preserved and used for re-vegetation;
- borrow areas are provided with gentle side slope that are re-vegetated and connected to the nearest drainage channel to avoid the formation of cess pools during the rainy season;
- control the disposal and ensure the vegetative stabilisation of spoil.

III. Climate

The proposed improvement/construction works will be localised activity and the Project should not have any significant impact on climatic conditions viz. rainfall, temperature and humidity in the project area.

IV. Surface & Ground Water Drainage & Hydrology

The natural courses of streams should, therefore, as far as possible be maintained. Appropriate temporary diversions of cross water-courses may be made if necessary. These streams should, however, be brought back to their natural course as soon as possible. Disposal of construction debris in streams and rivers should be avoided. This would disturb the water flow, cause siltation and also make the area look unattractive.

Minor impacts on water resources are expected during the construction phase. The rehabilitation of existing bridges may also cause soil erosion and turbidity in immediate downstream water bodies. To mitigate this, water-bodies slope stabilities should be monitored and, if necessary, appropriate remedial measures applied throughout the construction period. If possible, construction work at bridges should be avoided during the rainy season. This should minimise erosion and sedimentation.

The likely impacts of surface water movements are changes in the natural drainage systems and downstream scour and erosion due to constriction in flows. If suspended solid concentrations in the water are affected, this could also affect aquatic ecology. Construction activities could also lead to:

Temporary pollution of water bodies: from spillage of chemicals and oil at construction sites and waste from construction camps;

Discharge of sediment-laden water from construction areas;

Uncontrolled surface water discharge over the road edge: creating large-scale erosion on down-slopes.

To mitigate these impacts the following should be implemented:

- Chemicals and oils should be stored in secure, impermeable containers, and disposed of well away from surface waters;
- No vehicle cleaning activity should be allowed within 300 m of water bodies/ drains;
- Construction camps should be equipped with sanitary latrines that do not pollute surface waters;
- Work on bridges and culverts should be limited to dry seasons, when many of the smaller streams will have low water - water diversion works can thus be minimised and the original course restored immediately after the work has been completed;
- Drivers should be made aware of diversions and other works at bridge construction sites; - this is particularly important at night, if accidents are to be avoided;
- Drainage structures should be properly designed to accommodate forecast discharges;
- Side drain waters must be discharged at every available stream crossing so that the amount of discharge water is minimised and minimal erosion occurs at the water outlets;
- Lined drainage structures should be provided;
- Where an increased discharge of surface water endangers the stability of the water outlet, erosion protection measures such as

bio-engineering measures, ripraps, check dams etc. should be incorporated;

- In areas with high water tables, seepage may occur and side drains and up-slope catch drains must always be lined - this so that no surface water can penetrate into the subsoil; and
- All debris and vegetation, clogging culverts should be regularly cleared.

Ground water pollution most frequently occurs when a disposal site contains chemical substances, which are leached out during precipitation and then percolate to the ground water table. It is envisaged that no such materials, in any substantial quantity, will be used in the project. Negative impacts on ground water quality are, therefore, not anticipated - neither during the construction nor operation phases.

V. Air Quality

During construction, and at the micro-level only, air quality may be degraded for short periods. This will occur from: (i) the exhaust emissions from the operation of construction machinery; (ii) fugitive emissions from brick, concrete, and asphalt plants; (iii) the dust generated from the haulage of materials, exposed soils and material stockpiles, (iv) cutting and filling of hill slope, (v) cleaning of the road, (vi) material loading, (vii) unloading and (viii) blasting activities. The impact is, however, expected to be localised, temporary and confined to construction areas.

Negative air quality impacts during construction are likely to result from three main sources, viz. (i) emissions from construction equipment, including delivery trucks; (ii) fugitive dust from earth-moving operations and demolition; and (iii) localised increased traffic congestion in construction areas.

The negative impacts on air quality during construction stage can be classified as in Table 5-8. There are two types of pollution, dust pollution and pollution from harmful gases.

Table 5-8: Impact on Air Quality during Construction Stage

S.No	Impact	Source
1	Generation of Dust (SPM)	• Transportation and tipping of cut material - while the former will occur over the entire stretch between the cutting location and disposal site, the latter is more location specific and more intense; • Blasting operations; • Activation of landslides and rock falls etc.; • Transportation of raw materials from quarries and borrow sites; • Stone crushing, handling and storage of aggregates in asphalt plants; • Site levelling, clearing of trees, materials loading/unloading at construction site, construction of bridges; • Concrete batching plants; • Hot mix plants – due to the mixing of aggregates with bitumen; and • Construction of structures and allied activities
2	Generation of polluting gases including SO ₂ , NO _x and HC	• Hot mix plants; • Large construction equipment, trucks and asphalt producing and paving equipment; • The movement of heavy machinery, oil tankers etc. on steep slopes will cause much higher emissions of gases; • Toxic gases released through the heating process during bitumen production; and • Inadequate vehicle maintenance and the use of adulterated fuel in vehicles.

On the proposed Anakapalle - Atchutapuram road section, it is expected that air quality will be affected to some minor extent by dust and particulate matters generated by construction, vehicular movements, site clearance, earth filling and material loading and unloading. The impacts are, however, expected to be localised, temporary and confined to construction areas. Care should, however, be taken at sensitive locations so that harmful impacts can be minimised.

As it is expected that Suspended Particulate Matter (SPM) levels will increase during construction, certain mitigation measures are suggested in order to bring down these levels to prescribed standards (or) as far as is possible.

The following actions should be implemented:

- There should be regular check-up and maintenance of construction equipment - the idling of engines should be discouraged and machinery causing excessive pollution (i.e. visible clouds of smoke) should be banned from sites;
- Hot mix plant, Wet mix plant & Concrete batching plant should be operated within the permissible limits prescribed by CPCB, and be located away from settlements;
- Contractor should submit a dust suppression and control programme to the APRDC prior to construction - this plan should detail actions to be taken to minimise dust generation and identify equipment to be used;
- Vehicles delivering loose and fine materials should be covered to reduce spills;
- Controlled blasting, if necessary, shall be carried out only with the prior approval of the APRDC;
- Bitumen emulsion should be used wherever feasible, and
- Bitumen heater should be used and the use of wood for fuel shall be prohibited.

VI. Noise Levels

The existing noise in the project area is not a problem and are within the permissible limits. During the construction period, noise will be generated from the operation of heavy machinery, the haulage of construction materials to the construction yard and the general activities at the yard itself. Concrete mixing and material movements will be primary noise generating activities and, most likely, will be uniformly distributed over the entire construction period. These construction activities are expected to produce noise levels in the range of 80 – 95 dB (A). Piling, if necessary, will also cause vibration. Noise and vibration from this source will be unavoidable but the impact will only be temporary and will only affect people living or working near piling locations. In construction sites within 500 metres of a settlement, noisy operations should cease between 22:00 and 06:00 hrs. Regular maintenance of construction vehicles and machinery must also be undertaken to reduce noise. The impact and sources of noise and vibration are summarised in

Table 5-9

Table 5-9: Likely Impact on Noise Quality in the Vicinity of Project Area

Impact	Source
Increased Noise levels causing discomfort to local residents, workers and local fauna	Mobilisation of heavy construction machinery; Accelerations/decelerations/gear changes – though the extent of impact will depend on the level of congestion and smoothness of the road surface; Excavation work for foundations; Construction of structures and other facilities; Crusher plants, Hot mix plants; and Loading, transportation and unloading of construction materials.

Typical noise levels associated with various construction activities and equipment's are presented in Table 5.4 below:

Table 5-10: Typical noise levels of principal construction equipment's (Noise Level in db (A) at 50 Feet)

Construction Equipment's		Construction Equipment's	
Bulldozer	80	Crane	75-77
Front end loader	72-84	Welding generator	71-82
Jack hammer	81-98	Concrete mixer	74-88
Crane with ball	75-87	Concrete pump	81-84
		Concrete vibrator	76
Excavation and Earth Moving		Air compressor	74-87
Bulldozer	80	Pneumatic tools	81-98
Backhoe	72-93	Bulldozer	80
Front end loader	72-84	Cement and dump trucks	83-94
Dump truck	83-94	Front end loader	72-84
Jack hammer	81-98	Dump truck	83-94
Scraper	80-93	Paver	86-88
Grading and Compaction		Landscaping and clean-up	
Grader	80-93	Bulldozer	80
Roller	73-75	Backhoe	72-93
		Truck	83-94

Construction Equipment's		Construction Equipment's	
Paving		Front and end loader	72-84
Paver	86-88	Dump truck	83-94
Truck	83-94	Paver	86-88
Tamper	74-77	Dump truck	83-94
Source: U.S. Environmental Protection Agency, noise from Construction Equipment and Operations. Building Equipment and Home Appliance. NJID. 300.1 (December 31, 1971)			

The noise levels indicated for various construction activities/equipment, while far exceeding permissible standards, will occur only intermittently and be only temporary. Despite this, these extremely high sound levels present a real risk to the health of workers on- site.

Since the anticipated post-construction volumes of traffic on the project road is expected to be low, impacts from higher noise levels will have little significance even though present noise levels in these areas are low. In these areas, the noise produced during construction will also not have a significant impact, if proper mitigation measures are taken. Mitigation Measures should include:

Construction machinery should be located away from settlements;

Careful planning of machinery operation and the scheduling of such operations can reduce noise levels. The use of equipment emitting noise not greater than 90 dB (A) for an eight-hour operations shift and, when possible, the siting of construction yards at least 500 metres from residential areas should be adhered to:

- Controlled blasting should only be carried out with prior approval from the Engineer in charge;
- Contractors should be required to fit noise shields on construction machinery and to provide earplugs to the operators of heavy machines;
- Blasting should be conducted only during day-light hours; and
- Only controlled blasting shall be conducted, if necessary.

VII. Topography and Appearance

Construction activities of the project roads will bring permanent changes in the local-level topography and appearance of the project site. There will be a loss in aesthetic beauty of the project area mainly due to the earthwork. However this is usually a temporary phenomenon limited to the construction stage and the stage immediately following it. Proper re-vegetation activities and natural resumption of site stabilization generally bring back the previous look of the area.

The following table elaborates potential effects on the topography and appearance and appropriate mitigation measures.

S.NO	Construction activity	Potential effect on topography and appearance	Mitigation
1	Stone quarrying	Scarring of landscape and potential landslides (rock slides/falls). There may be permanent changes in the landscape.	Stone quarrying should only be undertaken in legally approved areas. Controlled and environmental friendly quarrying should be carried out to minimise landslides and erosion
2	Earthwork from borrow areas	Scarring of landscape due to unearthing activities, minor but permanent changes in landscape	Borrow areas should be in legally approved locations. As soon as construction activities are complete, they should be re-vegetated and brought back as far as possible to their previous appearance.
3	Waste disposal	Disposal of cut soils and debris at improper locations such as hillside below the road will make the area look untidy and unattractive.	Cut off material should be used to widen the road or disposed of at proper disposal sites.
4	Establishment of labour camps	Disposal of waste and litter at improper locations and deforestation for fire-wood will make the area look dirty and unattractive.	Provision and allocation of proper waste disposal bins and sites are required. A supply of cooking gas should be provided by the contractor to eliminate the use of fire wood.

5.6 Ecological Resources

I. Wildlife

There is no forest area in the Project Influence area nor have any wild-life crossings been reported by the Forest Department.

II. Vegetation

The project road passes through plain terrain. Removal of the existing road-side vegetative cover and the uprooting of about 4251 trees which will reduce the ecological balance in the areas. This will enhance soil erosion. Another impact from road construction activities and deriving from quarrying, preparation and transfer of stone aggregates and earthwork, is the accumulation of dust on the surrounding vegetation. This will lead to deterioration of the vegetative health, which in turn will affect the ecology as well as the aesthetic beauty of the area. Induced impacts may result from the following

- Increased earth and rock extraction;
- Construction crew demands for wood as a fuel and for building materials;
- Construction crew demands for food;
- To minimise negative impacts on the vegetative cover the contract documents should specify that;
- All wood building material for workers' housing should be brought from outside the project area;
- Workers should be supplied with non-wood fuels such as kerosene or liquefied petroleum gas for the duration of the contract;
- All contractor equipment and plants should be cleaned to the satisfaction of the project Engineer-in-charge prior to their relocation to project sites;
- During site clearance, care should be taken to ensure that the minimum area of vegetation is affected;
- Adequate Water bowser shall be used for sprinkling of water during construction activity on haul roads, so that dust deposition problem on vegetation are minimised.

It is estimated that about 1737 trees will need to be removed for widening of the road. Generally only part of the removed vegetation can be revived. Revival can be maximised by applying the following methods:

- Clearing only the necessary amount of vegetation from the project sites;
- Protection of slopes of road embankment by turfing with Kew grass (or) locally viable species.

5.7 Human Resources

I. Health, Safety and Hygiene for Construction Workers

Construction of the road will inevitably result in the generation of waste. In isolated places, the amount of waste generated may be greater than

normal because of substandard subsoil materials, which will need to be replaced.

The Contractor will be required to control the construction site, keep it clean and provide facilities such as dust bins and dust collectors for the temporary storage of all waste. This waste should be stored adequately to avoid the pollution of water supplies and water sources and to avoid dust formation. The Contractor should be responsible for the safe removal and/or storage of all waste in order to prevent environmental pollution of any type that may be harmful to people or animals.

All necessary safeguards should be taken to ensure the safety, welfare and good health of all persons entitled to be on the sites and to ensure that works are carried out in a safe and efficient manner. The personnel working at vulnerable site locations should wear safety helmets and strong footwear. It should be ensured that all workmen and staff employed on site use proper safety equipment, for example, eye protectors, ear plugs, safety helmets, the designated safety equipment when working over water and that proper rescue equipment is available. Fire extinguishers and first-aid equipment's shall be kept at all sites.

Inevitably, injuries may occur. At the construction camps and at all workplaces, first aid equipment and nursing staff should, therefore, be available.

The construction camps are anticipated to house up to 200 people for about eighteen months. With this concentration of people, the potential for the transmission of diseases and illnesses will increase.

The main health and safety risks during construction will arise from: inadequate sanitation facilities in worker camps; introduction of sexually transmitted and other diseases, by immigrant workers; and outbreaks of malaria, typhoid, cholera etc. amongst the labour force. The following actions should be undertaken at construction camps and stipulated in construction contracts: the contractor should be required to submit and obtain approval for a health and safety plan prior to the commencement of work; there should be provision of adequate health care facilities; and workers should be required to undergo pre-employment medical screening and treatment (if required) and periodic health checks thereafter.

Additionally, the project should support a public health education programme for workers and villagers covering road safety, malaria, hygiene, and sexually transmitted diseases. The district health departments should participate in monitoring and education of communities and workers affected by the project.

II. Nuisance to nearby properties

Nuisance to nearby properties is likely to result from: noise and vibration from mechanical devices and construction plant; dust during quarrying, construction and the trafficking of new surfaces prior to sealing; gaseous emissions from heavy equipment; and fumes from asphalt boiling sites.

Presently air/dust pollution is not a major issue in these areas. Nonetheless, there should be regular watering of the road surfaces or the application of emulsion coats near villages, where dust is a nuisance. Noise generating equipment such as power generators and concrete mixers should be kept away from populated/commercial areas. Provisions should also be incorporated into the contract to require the use of dust suppression measures.

III. Interference with Utilities and Traffic

On the project road, utilities interfere with the RoW at few locations. There are, however, few locations of electricity and telephone cables that will have to be moved prior to construction. This should not be a major problem.

Traffic may experience minor delays when diverted around active construction areas, but will be more severely hampered at the locations where temporary road closures are necessary. Such may, however, be unavoidable. Danger points should have proper signs indicating the nature of the problem envisaged. All signs should be distinct and visible.

IV. Community Impacts

There will be minor widening work on entire road length - in which case the resettlement problem will be only minor or nil.

A public consultation process has been undertaken as part of the IEE study.

Construction camps may, however, put stress on local resources and the infrastructure in nearby communities. In addition, local people may raise construction-process related grievances with the workers. This sometimes leads to aggression between residents and migrant workers. To prevent such problems, the Contractor should provide the construction camps with facilities such as health care clinics, places of worship and occasional entertainment. The use of local labourers during the construction will, of course, increase benefits to local peoples and minimise these problems.

Construction activity will, however, inevitably impact community resources, often adversely. Religious sites and places of public entertainment are particularly vulnerable, though the owners of the latter will probably welcome the increased, albeit temporary, patronage.

V. Quality of Life

The impact of the improvements of project road on the socio-economic environment will be significantly beneficial. Improved access and reduced travel times and costs will be major stimuli to economic growth, particularly in rural areas. The better access of agricultural goods to market will be particularly important and a major contributor to poverty reduction.

Increased labour mobility will also occur. This may, however, have both positive and negative impacts. Increased access is a two-way phenomenon, and the corollary to increased access to the project areas is increased access for the residents of these areas to more urban life-styles. Out-migration may, in fact, be the principal outcome. There is also the likelihood of the relocation of homes and businesses to new road-side locations.

During construction, benefits to local people can be maximised if the Contractor recruits construction workers locally. Where possible, he/she should also not discriminate in the employment of women.

The long-term effects of these roads on poverty reduction are, consequently, expected to be significantly positive.

VI. Construction Materials

The use of proper sources for stone and aggregates has become a major issue in most states. Sand and gravel are often obtained from river deposits. Jurisdiction over stone and aggregates is shared between the "Geological Survey of India" and the "State Forest Department". The "Geological Survey of India" issues licences for major mineral developments. The "Forest Department" issues permits for stone quarrying and for sand and gravel extraction. This is largely because these are mostly found on forest lands. Roadside quarrying is officially discouraged, but unofficially continues, invariably by petty contractors.

Road maintenance, repair and new construction will continue to cause large demands for construction materials. There is a clear need for a better materials supply policy in each district, both to minimise environmental impacts of small-scale, poorly managed operations and to improve the quality and reliability of supply. In some districts, it may be appropriate to develop centralised quarries, if an operator can be attracted. In any case, pre-designation of sources would give Contractors a level playing field for bidding and minimise incentives for environmentally damaging cost cutting.

As mandatory requirements for clearance of project every quarry and borrow areas: should be subjected to a site specific environmental investigation; should work according to an approved plan; and should be left in a safe condition or restored to a productive land use. Subject to these conditions, obtaining construction materials for projects will not cause unacceptable impacts.

Quarry and borrow pits may be filled with rejected construction waste and afterwards should be given a vegetative cover. If this is not possible, then the excavated slopes should be filled in such a way that they resemble an original ground surface.

5.8 Mitigation for Quarries:

Aggregates should be sourced only from licensed quarry sites that comply with environmental and other applicable regulations Occupational safety procedures/practices for the work force and as per law should be adhered to in all quarries Quarry and crushing units should be provided with adequate dust suppression measures - there should, for example, be water sprinkling in work areas and along the approach roads to quarry sites; There should be regular monitoring of the quarries by concerned authorities - such to ensure compliance with environmental management and monitoring measures.

5.9 Mitigation of Borrow Areas:

Prior approval should be obtained from concerned authorities and all local environmental regulations be complied with; Within all identified borrow areas, the actual extent of area to be excavated should be demarcated with signs and access to the operational area controlled; Borrow pit plant and machinery should conform to EPA noise emission regulations Protective gear, for example earplugs, should be provided to the workforce exposed to noise levels beyond threshold limits and there should be proper rotation of such personnel; and All operation areas should be water sprinkled to control dust levels to national ambient air quality standards. The project will require large amounts of bitumen or bitumen emulsion. This will, most likely, be supplied in drums. These empty bitumen drums are generally recycled as steel sheeting, or used in road construction as parapets or for bank stabilisation. When supplied and used in this manner, bitumen is not regarded as a significant environmental hazard.

The project will require the import, transport and use of fuel and oils. Minor diesel spills are common in region, especially around fuel stations. The project provides an opportunity to assist the APRDC and Contractors in improving fuel handling practices so as to minimise future fuel spillage.

5.10 Environmental Effects Related to Operation

Noise Vibration, Air Pollution & Runoff

The current traffic flows along the project road is expected to increase because of improved economic activities associated with better access. The larger numbers of vehicles will be an additional source of noise and gaseous emissions. Repairs to culverts and new drainage work will eliminate/reduce the soil erosion problems presently caused by poor cross drainage.

Land Use and Settlements

The likely impacts on land use and settlement patterns are limited. Improved access will inevitably lead to increase in and out migration, but this is likely to occur gradually and over a prolonged period. There will be time for new residential areas to be established. There may, however, be a need to control ribbon development. The improvement of road will attract the new development along the road. However, this is not expected much in the near future due to topographic features, but still in long term induced development is expected.

Social Impacts

The likely social impacts can be divided into benefits that occur to local society and other benefits, in terms of improvements to the wider quality-of-life. Specific benefits to local people will include:

- Easier communication;
- Easier access to markets (both internally and regionally) with savings in travel times and costs;
- Enhanced market efficiency through better distribution and accelerated deliveries etc.,
- Improved access to health, education and other social services;
- Employment generation;
- Improved technical skills; and
- Enhanced economic activity.

Likely adverse social impacts will include

- Increased chances of exposure to communicable diseases, particularly during construction;
- Influxes of new settlers leading to increased pressure on natural resources causing hardship to local communities relying on local/forest resources; and
- Rural-to-urban migration causing labour shortages in the depleted rural areas and other negative impacts in the urban areas.

5.11 Potential Environmental Enhancement /Protection Measures

In order to improve the environment conditions the following measures should be taken during construction.

Sanitation and House Keeping at the Labour /Construction Camps

a) Site Selection

- Construction camps should be located at 200 - 500 m away from existing habitations - the living accommodation and ancillary

facilities for labourers should be erected and maintained to standards and scales approved by the Engineer-in-Charge;

- All sites used for camps should be adequately drained and they should not be subjected to periodic flooding; and
- Camps should be located such that drainage from and through the camps will not endanger any domestic or public water supply.

b) Water Supply

- An adequate and convenient water supply, approved by the appropriate health authority, should be provided in each camp for drinking, cooking, bathing and laundry purposes;
- Potable water supply systems for labour camp occupants should be as per the design approved by local public health/engineering departments and meet the water quality standards as prescribed by the WHO. In addition, the design of water system facilities should be based on the suppliers and the Engineer-in-Charge's estimates of water demands;
- Drinking water system must be monitored in accordance with water quality parameters as prescribed by the WHO. Any water supply system used for cooking purposes that is drained seasonally should be cleaned, flushed, and disinfected prior to use. Furthermore, a water sample of satisfactory bacteriologic quality, i.e. a sample showing not more than one coliform bacteria per 100 ml sample, should be obtained before such systems are placed into service; and
- All construction camps and other workplaces, good and sufficient water supply should be maintained to eliminate the chances of waterborne/water-related/water-based diseases and to ensure the health and hygiene of the workers.

c) Toilet Facilities and Hygiene

- Within the precinct of every workplace and accommodation unit, latrines and urinals should be provided in an accessible place, as per standards set locally. Where flushing systems cannot be provided, all latrines should be provided with a dry-earth system, which should be cleaned at least four times daily and at least twice during working hours and kept in a strict sanitary condition. Receptacles should be tarred inside and outside at least once a year;
- Toilet facilities adequate for the capacity of the camp should be provided - each toilet room should be located so as to be accessible;

- A toilet room should be located within 200 feet of the door of each sleeping room and no toilet should be closer than 100 feet to any sleeping room, lunch area or kitchen;
- An adequate number of urinals should be provided.

d) Waste Disposal

- The sewage system for the camps should be designed, built and operated to the satisfaction of the concerned public health engineering department, so that no health hazard occurs and no pollution to the air, ground or adjacent watercourse takes place - compliance with the relevant legislation must be strictly adhered to;
- Garbage bins should be provided in the camps and regularly emptied and the garbage disposed off in a hygienic manner to the satisfaction of relevant norms; and
- On completion of the works, all such temporary structures should be cleared away, all rubbish burnt, excreta tank and other disposal pits or trenches filled in and effectively sealed off and the outline site left clean and tidy to the entire satisfaction of the Engineer-in-Charge.

e) First Aid

- Injuries may inevitably occur and it is, therefore, pertinent to provide first aid facilities for all workers - at construction camps and at all other workplaces first aid equipment should be provided;
- Adequate transport facilities for moving injured persons to the nearest hospital should be provided and these should be in a ready-to-move condition;
- First-aid units should have an adequate supply of sterilised dressing material and should contain an appropriate selection of other necessary appliances.

f) Maintenance

- All buildings, rooms and equipment and the grounds surrounding them should be maintained in a clean and operable condition and should be protected from rubbish accumulation;
- All necessary means should be employed to eliminate and control any infestations of insects and rodents within all parts of the labour camps - this should include approved screening or other control of outside openings in structures intended for occupancy or food service facilities;

- Each structure made available for occupancy should be of sound construction, should assure adequate protection against the weather, and should include essential facilities to permit maintenance in a clean and operable condition – the comfort and safety of occupants should be provided by adequate heating, lighting, ventilation or insulation and, when necessary, the facilities to reduce excessive heat; and
- Each structure made available for occupancy should comply with the requirements of local building codes – though this should not apply to tent camps.

Guidelines for identification of Debris Disposal Sites

The locations of dump sites should be selected in such a manner that:

- Residential areas are not located downwind of these locations;
- Dump sites do not contaminate any water sources, rivers etc.;
- Dump sites have capacity at least equal to the amount of debris to be generated;
- Public perceptions about the manner and location of debris disposal are obtained and taken into consideration before finalising locations; and
- Permission from the head of the village/community is obtained.

Precautions to be adopted during Dumping of Debris/Waste Material

The contractor shall take the following precautions while disposing off waste material:

- During site clearance and disposal of debris, the contractor should take full care to ensure that public or private properties are not damaged/affected and that traffic is not interrupted;
- The Contractor should dispose off debris only to identified places or with prior permission of the Engineer-in-Charge of works;
- The Contractor should only dispose of debris from the improvement of public utilities after the proper consent of villagers and approval of the Engineer-in-Charge of works;
- In the event of any spoil or debris from the sites being deposited on any adjacent land, the contractor should immediately remove all such spoil debris and restore the affected area to its original state to the satisfaction of the Engineer-in-Charge of works;
- The contractor should at all times ensure that all stream/river courses and drains within and adjacent to the site are kept safe and free from any debris;

- When dust is likely to be created, and to dampen stored materials during dry and windy weather, the contractor should apply effective water sprays – this should also occur during the delivery and handling of materials;
- Materials having the potential to produce dust should not be loaded to a level higher than the side and tail boards of a truck and should be covered with a tarpaulin in good condition;
- Any diversions required for traffic during the disposal of debris should be provided with traffic control signals and barriers and after discussions with local people and the permission of the Engineer-in-Charge of works;
- During debris disposal, the Contractor should take care of surrounding features and avoid any damage to them;
- While disposing debris/waste material, the contractor should take into account the wind direction and location of settlements to ensure that any dust problems are avoided; and
- Adequate arrangements should be made to ensure that debris/waste material is disposed of to the nearest designated dumping site - reports on this activity should be provided regularly by NGOs/ Village Society Heads.

5.12 Rehabilitation Plan for Quarry/Borrow Pits

The following are important:

- The objective of the rehabilitation programme should be to make the quarry/borrow pit sites safe and secure areas. Securing borrow pits/quarry sites in a stable condition should be a fundamental requirement of the rehabilitation process. This can be achieved by filling the quarry/borrow pit floor to approximately the access road level;
- From the outset there should be a restoration plan, which should influence all day-to-day quarrying activities; and
- Land-use planning considerations should be taken into account when defining devising and finalising restoration works and these should take into account the need to preserve the environment and, if possible, to generate income for local communities. Other criteria which should be followed when rehabilitating quarry/borrow pits are-
- Quarries and borrow pits should be used to dispose of rejected construction waste and covered with a vegetation. If this is not possible, the excavated slopes should be smoothed and depressions filled in such a way that they look, more or less, like an original ground surface;

- During the execution of works, the Contractor should ensure: the preservation of trees and stock-piling of materials; the spreading of stripped material to facilitate water percolation and to allow natural vegetation growth; the reestablishment of previous natural drainage flows; the improvement of site appearance; the digging of ditches to collect runoff; and the maintenance of roadways. Once works are complete, the contractor should restore the environment around the work site to its original state;
- Immediate surroundings of quarry or borrow pits should be developed as a low maintenance reserve, with significant areas of native trees and shrubs and areas of longer grass forming open spaces.

6 Environmental Management & Monitoring Plan and Institutional Assessment

The management and monitoring mechanism required to ensure that the project implementation is carried out in accordance with the due regards to environment. These includes major environmental issues and associated impacts, suggested mitigation measures, implementation and supervising responsibilities, a monitoring plan and institutional assessment and training requirements for successful implementation of the mitigation measures.

Environmental Management and Monitoring Plan

In the previous chapters we have dealt with the various environmental impact and mitigation measures associated with the road development. Adaptation of appropriate mitigation measures during design, construction and operation stages shall reduce the negative impacts of the project to acceptable limits.

Keeping in view various environmental issues associated with road section improvement, an environmental impacts and mitigation matrix including time frame and the implementing responsibilities has been worked out and presented in Table 6.1. A common Impact and Mitigation matrix applicable to road project is provided for various stages of project. Site specific impacts and mitigation measures at construction stage are worked out and presented separately for proposed road section i.e. Anakapalle - Atchutapuram Road.

Environmental monitoring plans form the basis for verifying the extent of compliance during the implementation and operation stages of the project.

The objectives of an environmental monitoring programme are:

- To evaluate the performance of mitigation measures proposed in IEE
- To provide information which could be used to verify predicted impacts and thus validate impact prediction techniques
- To suggest improvement in environmental mitigation measures if required
- To provide information on unanticipated adverse impacts or sudden change in impact trends. The Table 6.2 gives a generic monitoring plan for the road sector development. However, depending on ground realities, these needs evaluation.

Table 6-11: Environmental Impact and Mitigation Matrix for Project Road

(Note: A generic impact and mitigation matrix applicable to all road projects is provided for various stages of project. Site specific impacts and mitigation measures at construction stage are addressed separately for proposed road section i.e. Anakapalle - Atchutapuram)

Project Stage/Activity	Potential Impacts	Mitigation Measures	Location	Responsibility during Implementation	Responsibility during Monitoring
1. General Matrix applicable to all road sections					
1.1 Preliminary Planning and Design Stage					
1.1.1 Widening options of project road	Location on agricultural land, dense forests, wildlife habitat, unstable sites and religious/ cultural sites. Change in widening option determined during the detailed design stage changing the scope or scale of environmental impacts predicted in the IEE.	Widening on other side of agricultural land, dense forests, wildlife habitat and unstable sites. Widening should avoid religious/cultural sites. Additional environment studies for new alignments (if required).	Entire project length	APRDC /Supervision Consultant	APRDC
1.1.2 Location of construction camps and	Location in inappropriate locations such as close	Construction camps should be located at least 500m away from community	Project construction sites	APRDC /Supervision	APRDC

Project Stage/Activity	Potential Impacts	Mitigation Measures	Location	Responsibility during Implementation	Responsibility during Monitoring
Contractor facilities	to the local communities, community drinking water source etc.	areas and away from water resources, and at least 1 km away from reserve / protected forest stretches. Village Forest Management Committees should be consulted before locating temporary project facilities.		Consultant	
1.1.3 Location of quarry sites	Location in un approved areas, forest areas etc.	Only government approved quarry sites should be planned for project use. Quarries should not be located in the locations of reserve / protected forest stretches.	Environmentally and technically suitable sites near the project road	APRDC /Supervision Consultant	APRDC /Department of Geology & Mines
1.1.4 Location of borrow pits	Location in unstable areas or close to village	Location in environmentally sound areas and away from villages Borrow areas should not be located in the locations of reserve / protected forest stretches.	Environmentally and technically suitable sites and near the project road	APRDC /Supervision Consultant	APRDC

Project Stage/Activity	Potential Impacts	Mitigation Measures	Location	Responsibility during Implementation	Responsibility during Monitoring
1.1.5 Obtainment of appropriate NOC/permits	Delays in processing permits causing further delay in initiation of project construction	Processing of NOC/permits on a timely basis.		APRDC /Supervision Consultant	ADB
1.1.6 Preparation of project detailed design and contract bidding documents	Exclusion of environmental management and mitigation measures hence lack of EFRC during construction	Incorporation of all mitigation measures into the project detailed design and contract bidding documents		APRDC /Design Consultant	ADB
1.7 Removal of encroachment /structures along the road.	Loss of livelihood and structures.	Compensation against loss of structures and should be rehabilitated as per the provisions of resettlement plan.	locations where resettlement required	APRDC /Supervision Consultant	ADB
1.2 Construction Stage					
1.2.1 Removal of vegetation and felling of about 4251 trees	Change in micro level habitat/environment. Soil erosion. Scarring of landscape.	Removal of only necessary vegetation. Re-vegetation immediately after earth removal activities. Compensatory afforestation	Area of the ROW mainly forests area	Contractor / Forest Department	APRDC

Project Stage/Activity	Potential Impacts	Mitigation Measures	Location	Responsibility during Implementation	Responsibility during Monitoring
		in the available land @ 1:2 ratio in consultation with State Forest Department			
1.2.2 Quarrying	Landslides (rockslides/falls), scarring of landscape	Use of environmental friendly techniques of quarrying.	Quarry sites	Contractor	APRDC
1.2.3 Crushing of stone and transport of stone and materials.	Dust pollution for construction labourers and local vegetation. Air pollution from machinery and vehicle exhaust. Noise pollution and disturbance to nearby wildlife.	Water sprinkling of stone crushing site. Proper coverage of vehicle transporting stone and material. Regular maintenance of machinery and vehicles. Conduction of stone crushing and transportation activities only during the day. Stone crushers should be located away from settlements and forest stretches.	Stone crushing sites and road construction	Contractor	APRDC /Mining department
1.2.4 Establishment of construction camps	Improper waste disposal. Loss of aesthetic beauty. Health issues.	Provision of cooking gas. Provision of proper waste disposal facilities as well as health facilities.	Construction camp sites	Contractor/APRDC	APRDC

Project Stage/Activity	Potential Impacts	Mitigation Measures	Location	Responsibility during Implementation	Responsibility during Monitoring
1.2.5 Operation of machinery and equipment and general activities of labourers	Spillage/leakage of chemicals and oil and contamination of soil and water resources. Injury to labourers. Respiratory problems from dust and machinery emissions. Hearing problem due to high level of noise. Traffic delays and congestion.	Proper storage and handling of chemicals and oil. Provision of adequate facilities such as construction hats, facemasks, earplugs, gloves etc. Provision of well-equipped First Aid Kit and health facilities. Preparation of traffic control plans, proper maintenance of road surface to allow smooth flow of traffic.	Construction site	Contractor	APRDC
1.2.6 Water supply for construction activities and labour camps	Disruption of water supply of local communities	Independent arrangements be made for requirements for construction work in such a way that water availability and supply to nearby communities remains unaffected	Construction camps and work place	Contractor	APRDC
1.2.7 Unexpected environmental	Destruction of constructed road and road furniture,	Undertake appropriate remedial actions in coordination with relevant	Construction camp and sites	Contractor, APRDC	APRDC

Project Stage/Activity	Potential Impacts	Mitigation Measures	Location	Responsibility during Implementation	Responsibility during Monitoring
problems/calamities (ex. flash floods) during construction activities	injuries, loss of lives and property.	government and local agencies.			
1.2.8 Implementation of EMP during construction.	Inadequacy of environmental management measures or deviation from the EMP measures mentioned in the contract documents.	Strictly following and implement the EMP measures mentioned in the contract documents and maintaining proper documentation of measures taken.	Construction camp and sites	Contractor	APRDC, ADB
1.3 Operation Stage					
1.3.1 Movement of vehicles	Air pollution from emissions. Noise and vibrations causing disturbance to residents and domestic animal Contamination of soil and water resources due to poor drainage.	Pollution under Control certificate for all vehicles and periodical noise monitoring of equipment's. Periodic cleaning of drains.	Along the road	Contractor, APRDC	APRDC /Local Traffic Police

Project Stage/Activity	Potential Impacts	Mitigation Measures	Location	Responsibility during Implementation	Responsibility during Monitoring
2. Site Specific impact matrix applicable to Anakapalle - Atchutapuram road section					
2.1 Construction Stage					
2.1.1 Earth removal from borrow areas	Scarring of landscape due to improper disposal of debris. Soil erosion. Disruption of local drainage. Siltation in nearby water bodies and hence negative effects on aquatic ecology.	Disposal of debris at proper disposal site. Proper re-vegetation of borrow areas. Provision of appropriate drainage structures/facilities.	Borrow area, water bodies.	Contractor	APRDC
2.1.2 Removal of vegetation and uprooting of trees (about 4251)	Change in micro level habitat/environment. Soil erosion. Scarring of landscape.	Removal of only necessary vegetation. Re-vegetation of the space available within RoW immediately after earth removal activities. Removal of trees should be compensated with planting new trees @ 1:2 ratio on available space along the	Entire project length with vegetation.	Contractor / Forest Department	APRDC

Project Stage/Activity	Potential Impacts	Mitigation Measures	Location	Responsibility during Implementation	Responsibility during Monitoring
		road.			
2.1.3 Construction of culverts and bridges	Disruption of local stream course and aquatic hydrology.	Construction during dry season. Provision of appropriate drainage facilities and stream diversion structures.	Bridge / culvert location i.e.	Contractor/ APRDC	APRDC
2.1.4 Relocation of cultural properties	Disturbance to religious sentiments of the local communities	Religious structures should be left undisturbed, shifting (if require) shall be done in full cooperation with local people.	Locations of temples/shrines at km.	Contractor/ APRDC / local community leaders	APRDC
2.1.5 Construction at sensitive locations	Increased noise level during construction and operation	Construction of suitable noise barriers at these locations	Locations of schools, religious properties etc.	Contractor/ APRDC	APRDC

Table 6-2: Environmental Monitoring Plan

Environmental Features	Aspect to be Monitored	Time and Frequency of Monitoring	Location	Responsible party
A. Physical Environment				
i) Air and Noise	Level of PM ₁₀ , PM _{2.5} and SO ₂ and NO _x Noise levels on dB (A) scale	Before commencement of any construction activities.	At selected locations	APRDC
		Once in every section while construction is ongoing.		
		Once after completion of construction activities.		
ii) Water Bodies	Concentration of sediments and presence of construction debris.	Before starting of construction activities.	Major water bodies	APRDC
		During construction activities in the vicinity of each water body.		
		Once after completion of construction activities.		
	pH, BOD, COD, DO, TDS, NO ₃ and Coliform	Same as above.		
	Length of line drainage structures constructed and strengthened.	During construction activities in the vicinity of each water body.	Full length of project road	APRDC
	Length of damaged or missing line drains.	Before starting of construction activities.	Full length of project road	APRDC
		Once after completion of construction activities.		
	Total number, type and	Before starting of construction	Full length of	APRDC

Environmental Features	Aspect to be Monitored	Time and Frequency of Monitoring	Location	Responsible party
	lengths of cross drainage structures including bridges constructed or strengthened	activities. Once a year during construction activities Once after completion of all construction activities.	project road	
	Number of weak cross drainage structures.	Before starting of construction activities. Once after completion of construction activities.	Full length of project road	APRDC
iii) Geology	Number of cases of illegal quarrying or mining	Once a year after completion of construction activities.	Entire project length.	APRDC, DoM&G
B. Ecological Resources				
i) Flora	Total area of vegetative cover	Before starting of construction activities. Once after completion of construction.	Entire project section	APRDC, Forest Department
	Total number of trees	Same as above.		
	Average tree density	Before starting of construction activities. During construction (once a year per section) Once after completion of construction activities and thereafter once every year for 5 to	Full length of project road	APRDC, Forest Department

Environmental Features	Aspect to be Monitored	Time and Frequency of Monitoring	Location	Responsible party
		10 years depending on budget availability.		
	Number of cases of illegal tree felling	Once a year during construction activities Once after completion of all construction activities.	Full length of project road	APRDC, Forest Department
C. Social Environment				
i) Health	Number of accidents amongst construction workers.	During construction activities.	All construction sites along project road	APRDC, Contractor, local health officials,
	Number of accidents due to moving traffic amongst local community members	Before starting construction activities. Once a year during construction activities. Once every year after completion of construction activities.	All villages along project road	APRDC, Contractor, local health officials,
ii) Travel time	Time taken to travel within each road section.	Before starting construction activities. After construction activities	Full length of project road	APRDC, Contractor
	Number and extent of travel delays	During construction activities (throughout the year).	Full length of project road	APRDC, Contractor,

6.1 Environment Management Budget

Most of the measures have been addressed as part of good engineering practices, the costs for which have been accounted for in the engineering/cost. All costs towards preconstruction clearances/permission will be borne by executing agency. These costs are indicative. The environmental budget for the various environmental management measures proposed under the project is presented in Table 6.3.

A total proposed budget amount of Rs.81,48,000 will be allocated for implementation of environment safeguards under the project.

Table 6-3: Estimated Environment Management Cost

S.No	Activity	Amount in Rs	To be included in Budget under	Remarks
1	Tree Cutting and Compensatory Afforestation activities			
	Compensatory afforestation (1:2 basis) by forestry department 3474 Total Trees contractor through VSS committee	6948000	APRDC-under VCIC project	No. Trees to be planted 3474 Nos (1:2 basis)@2000/tree including maintenance for 5 Years Contractor to assign the VSS Committee for carrying out the additional 1:2 plantations.
2	Studies and Monitoring activities			
	Monitoring (air, water, noise, soil)	7,00,000	BOQ of Civil Works Cost	
3	Training	5,00,000	APRDC under VCIC Project	
	Total	81,48,000		

6.2 Institutional Assessment

The proposed Visakhapatnam-Chennai Industrial Corridor Connectivity Project is initiated and is being carried out by the Andhra Pradesh Road Development Corporation, Government of Andhra Pradesh. The responsibility for effective implementation of all project activities shall be with the Andhra Pradesh Road Development Corporation (APRDC).

APRDC has established an Environmental & Social Management Unit in its organisational setup to look after environmental and social concerns of road development projects. This unit is headed by the Managing Director and supported by the Project Director along with the Environmental Officer with support of Assistant Environmental Engineer. The existing capacity of APRDC in environmental management is adequate.

Additional assistance, to strengthen the environmental capacity of the APRDC might be useful in the form of training in environmental & social issues and long term monitoring.

A mitigation plan will be included in the Construction contract and the Contractor will be responsible for its implementation. The APRDC will monitor the implementation of these mitigation measures by the Contractors through Environmental Officer and its appointed Consultant at the site.

The APRDC will ensure that the project complies with the best environmental practices and meets the mitigation and monitoring requirements described in the environmental report.

During construction, the likely impacts will also be required to be monitored. The likely changes in the quality of various parameters such as air quality, water quality and noise levels shall be monitored periodically - say quarterly during seasonal changes.

6.3 Training

Need for Training

The APRDC's present environmental and social expertise shall be strengthened in order to enable the APRDC to integrate social and environmental issues into its day-to-day operations and Internalising environmental and social issues in its future road development projects.

Targets of Proposed Training

The training programme should Expose senior members (CE, SE, and EE) of the APRDC to environmental and social issues associated with road projects and train and equip members of the Divisional office to implement and supervise environmental mitigation measures.

The senior members of the Environmental unit might then be given the additional responsibility for active dissemination of the culture of environmental/ social consciousness and ethics with other members.

In order to disseminate environmental experiences gained, each staff member would be required to maintain good records and prepare dissemination notes on specific issues and problems encountered and resolved, and how the experience gained could be integrated into future road projects.

6.4 Institutional / Capacity Building

To enhance the capacity of officials for effective implementation of proposed mitigation measures and monitoring the resultant effects, as well as create awareness amongst workers and public, the training and awareness programme is planned.

The institutions/agencies like regional office of MoEF, SPCB/CPCB, and Indian Institute of Technologies can be consulted for such trainings. Independent subject's experts/consultants (e.g., for the environmental awareness program, impact assessment specialist will be the resource person) can also be the resource persons to impart trainings. These experts/agencies shall be appointed based on specific need for the training. A separate budget for training can be adopted under VCIC -APRDC of CSC budget.

S.No	Target group	Subjects	Method	Time Frame
1	All staffs of APRDC including PIU project staff involved in implementation of the project	Environmental Overview: Environmental Regulations, sub-project related provisions of various Acts/ Guideline Procedures of EC and FC, process and methodology for IEE, EMPs	Lectures cum interaction	Before beginning of the implementation of the subproject.
2	Managers (Env) at PIU, Supervision Consultant's Environmental Specialists and Select NGOs	Implementation of EMPs: Basic features of an EMP, Planning, designing and execution of environmental mitigation and enhancement measures, monitoring and evaluation of environmental conditions – during construction and operation	Workshops and Seminars	Before the construction begins
3	Environmental officer, design team, Supervision Consultant Construction Contractors' staff	Environmentally Sound Construction Practices: Clean construction technology, alternative materials and techniques for construction, Waste Management and minimization in construction, pollution control devices and methods for construction sites and equipment, Environmental clauses in contract documents and their implications, protection of flora and fauna Environmental monitoring during construction	Workshops and Site visits	Before the construction
4	PIU and Supervision Consultant, NGOs and community representatives	Monitoring Environmental Performance during Construction: Air, Water, Soil and Noise, tree survival Monitoring requirement and techniques, Evaluation and Review of results, Performance indicators and their applicability possible corrective actions, reporting requirements and mechanisms	Lectures, Workshop and site visits	During initial phases of construction
5	PIU and Supervision Consultant, NGOs and community representatives	Long-term Environmental Issues in Project Management: Designing and implementing environmental surveys for ambient air, noise, biological and	Workshops and seminars	During Implementation of the Subproject

S.No	Target group	Subjects	Method	Time Frame
		water quality surveys, data storage, retrieval and analysis, contract documents and environmental clauses, risk assessment and management, contingency planning and management and value addition		
6	Public /contractors workers	Awareness programmes on environmental protection and measures being implemented by APRDC and their role in sustaining the measures taken including for noise pollution, air pollution, safety, soil conservation, and agricultural productivity enhancement	Workshops	During construction and initial phase say 3 years of operation
7	APRDC Staff, Supervision Consultant, Engineering Staff of Contractor.	Restoration of sites viz borrow areas, Construction Camps, Crushing units, HMP etc. And Reporting Formats/procedure	Lecture / Presentations	Before Contractor Demobilization

7 Grievance Redress Mechanism

All the three parties involved in this project implementation i.e. Contractor, CSC and executing agency will maintain complaint registers at their following respective offices Contractor's main site offices i.e. office of the Project Manager, CSC's main site office i.e. office of the Engineer's Representative and PIU office .

Level 1 – PIU level: All public complaints regarding environmental issues received by any of the above mentioned offices will be entered into the register with specific details such as name and address of the person or representative of the community registering a complaint, the details of complaint, and time. The Executive Engineer and Engineer's Representative will immediately communicate the details of the complaint to the Contractor.

The environment and safety officer of the contractor will promptly investigate and review the environmental complaint and implement appropriate corrective actions to arrest or mitigate the cause of the complaints within 3 days time of receiving the complaint.

The contractor will report to CSC environment expert about the action taken on the complaint, also within 3 days time of receiving the complaint, for his further intimation to EE PIU. The person making the complaint will also be intimated by the complaint receiving person or his representative, about the action taken, within 3 days.

Level 2 – State level: Grievances not redressed by the PIU level will be brought to the State level Grievance Redress Committee (GRC). The State level GRC will be headed and chaired by Project Director (Superintendent Engineer). The state level GRC will comprise of the following:

- I. Member (Administration), APRDC
- II. Environmental Specialist, CSC
- III. A representative from the respective local community or NGO representative
- IV. Representative of concerned agency such as Forestry Department or State
- V. Pollution Control Board depending on the nature of the complaint/issue.

The main responsibilities of the GRC will be to: (i) record grievances, categorize, and prioritize grievances and resolve them as soon as possible; (ii) immediately inform the EA of serious cases; and (iii) report to complainants on decisions made regarding their grievances.

Within three weeks of receiving the grievance from the PIU level. The decision must include the timeline for addressing the grievance. Grievances related to resettlement benefits, compensation, relocation, replacement cost and other assistance will be addressed by following the grievance redress system provided in the RP.

8 Public Consultation and Information Disclosure

8.1 Meaningful consultation

Meaningful consultations were carried out during detailed design and IEE preparation. All the five principles of information dissemination, information solicitation, integration, coordination, and engagement into dialogue were incorporated in the consultation process. A framework of mitigating different environmental impacts likely from the project was strengthened and modified based on opinions of all those consulted, especially at the micro level by setting up a dialogue with the village people from whom information on site facts and prevailing conditions were collected. This will be continued during the implementation of the project through grievance redress mechanism.

8.2 Objectives of the Public Consultations

Public consultations were held to allow the incorporation of relevant views of the stakeholders in the final project design, mitigation measures, implementation issues, and enhance the distribution of benefits. Stakeholder's consultations were held with intent to understand their concerns, apprehensions, overall opinion and solicit recommendations to improve project design and implementation. Informal meetings, interviews were organized covering the entire project design stage. Consultations provide affected public a platform to ensure incorporation of their concerns in the decision making process and foster co-operation among officers of APRDC, the community and the stakeholders to achieve a cordial working relationship for smooth implementation of the project. It inculcates the sense of belongingness in the public about the project.

The discussions were designed to receive maximum inputs from the participants regarding their acceptability and environmental concerns arising out of the sub-project. They were given the brief outline of the project to which their opinions was sought particularly in identifying and mitigating any potential adverse impact.

8.3 Methodology

Consultation with the stakeholders, beneficiaries, and community leaders were carried out using standard structured questionnaires as well as unstructured questionnaires. Questionnaire survey/discussions were designed to obtain background information and details of general environmental issues that concern people in the project area. In addition, environmental issues were discussed with relevant organizations, government officials, beneficiaries, community leaders and experts. In addition, personal discussions with officials, on site discussion with affected stakeholders, and reconnaissance visits have also been made to the project area.

8.4 Project Stakeholders

All types of stakeholders were identified to ensure as wide coverage as possible like Residents, shopkeepers and businesspeople who live and work along the road specially the project affected persons, road users/commuters, executing agency, government institutions whose remit includes areas or issues affected by the project state environment and forest department, Pollution Control Board (PCB), Irrigation Department, Public Health Engineering (PHED) Department and most importantly the beneficiary community in general.

8.5 Consultations with Local People/Beneficiaries

The informal consultation generally started with explaining the project, followed by an explanation to potential impacts. Participant's views were gathered with regard to all aspects of the environment which may have direct or indirect impact on local people.

Key issues discussed were:

- Awareness and extent of the project and development components
- Benefits of the project for the economic and social upliftment of community
- Labour availability in the project area or requirement of outside labour
- Local disturbances due to project construction work
- Necessity of tree felling etc. at project sites
- Impact on water bodies, water logging and drainage problem if any
- Environment and health
- Flora and fauna of the project area
- Socio-economic standing of the local people and

Consultations were held at prominent villages where the more number of settlements are affecting. At these two locations the nearest vicinity villagers were presented, participated in the public consultation meetings.

Local community welcomed the decision of road widening and improvement proposal. They perceived several benefits like faster and cheaper connectivity, improved accessibility to better infrastructure facilities, reduction in migration, increased economic activities and appreciation in value of land and many others. But at the same time they apprehended that the risk of accident, air and noise pollution will increase due to high traffic density after widening.

Main demand and suggestions made by the participants are:

- Adequate compensation and rehabilitation assistance to affected households
- House sites for Relocation families
- Road safety measures
- Extensive plantation
- Protection of common properties such as Schools, Temples, Graveyards etc.,
- Restriction on honking near built-up areas and sensitive receptors
- Lighting in built-up areas and sensitive receptors
- Measures to minimize air and noise pollution
- Providing health and hospital facility at the junctions & crowded locations
- Bus Shelters construction
- Parking areas in markets and truck lay-byes near industries

Design considerations have been made to incorporate most of the suggestions and demands of the local people.

8.6 Consultations with Women and Vulnerable Groups

Further consultations with only women and vulnerable households (female headed households, households below poverty line etc.) were conducted as part of the social safeguards studies. The purpose of these exclusive discussions was to ensure women were aware about the project and understand their concerns and expected benefits out of the project.

There were various concerns that were raised by the women during the consultations. The women expressed a number of both key benefits and concerns that they perceive out of this subproject. The improvement of the road network will have positive impact as it will increase the frequency and quality of the transportation which will not only improve the accessibility issue but will also increase the value of land. They were also of opinion that the augmentation of the road network would help in creating employment opportunities for the local people.

However, the women participants did voiced their concerns regarding the safety of them and their children as they were of opinion that the widening of the road would increase the frequency of the vehicles which would lead to the risk regarding accidents. They were informed that adequate provisions for road safety and have been integrated in the road design by the technical design team to address the accident risks. The other negative impacts that they raised was the increase in the level of air and noise pollution as a result of the project. In response they were

informed that air and noise issues will be minimal. And necessary measures to reduce noise levels such as speed control, tree plantation and noise barriers will be installed in locations with sensitive receptors. Further details on the discussions held with women are provided in the RAPs.

8.7 Disclosure of information

IEE report will be made available at APRDC HQ and respective PIUs or Executive Engineer's offices. The same will be posted on APRDCs website. Based on ADB disclosure requirements, it will be posted on its website.

Note: The detailed information on Public Consultations is being furnished with Draft Resettlement Plan Report.

Table 8-1: List of Affected CPR's in the Anakapalle to Atchutapuram Road

Asset No	Chainage location	Name of The Village	Side	Distance from the Center line in Mts	Impact	Affected Property
1	2380	Nagulapally	L	12.5	Minor	Shiva Temple
2	2400	Nagulapally	R	11	Major	Devi Perantalu Kotalu
3	2645	Nagulapally	L	5	Major	Satthamma Temple
4	2790	Nagulapally	R	7.5	Major	Sri Kasi Anapurneswari Temple
5	2840	Nagulapally	L	8.5	Major	Human Temple
6	2850	Nagulapally	L	10	Major	Bus-stand
7	2900	Nagulapally	R	12	Major	Z.P.P.S School
8	3120	Nagulapally	R	6.5	Major	Vinayaka Temple
9	3820	Ompolu	R	6.5	Major	Pydimamba & Vinayaka Temple
10	4740	Munagapaka	L	8	Major	Pasara Maramma Thalli Temple
11	5530	Munagapaka	R	13.5	Minor	Church Compound
12	5590	Munagapaka	R	6	Major	MusiliThalli Temple
13	5635	Munagapaka	R	8.5	Major	Human Temple
14	5650	Munagapaka	R	5.7	Major	Bus-stand
15	5680	Munagapaka	R	13	Minor	Pump House

Asset No	Chainage location	Name of The Village	Side	Distance from the Center line in Mts	Impact	Affected Property
16	5700	Munagapaka	R	5	Major	Shivalam
17	6240	Munagapaka	R	11.3	Major	Maridamma Thalli Temple
18	7430	Munagapaka	R	6	Major	Maridamba Thalli Temple
19	7720	Munagapaka	L	7	Major	Anajaneya Swami Temple
20	7815	Munagapaka	R	3.2	Major	Mutyalamma Temple
21	8260	Munagapaka	R	5	Major	Mamidi Mahalakshmi
22	8635	Kondla karla	L	8.5	Major	Yellaramma Temple
23	8640	Kondla karla	L	13.5	Minor	Vinayaka Temple
24	8670	Kondla karla	R	6.5	Major	Hunuman Temple
25	12955	Chodapalli	R	3	Major	Chodamambica Temple
26	13410	Chodapalli	R	8	Major	Paradesimamba Pucca
27	13415	Chodapalli	L	10.8	Major	Paradesimamba Pucca
28	13705	Chodapalli	R	3.8	Major	Paradesimamba Semi Pucca
29	14105	Mosayyapeta	R	12.3	Minor	School Compound Walls& Toilets
30	14150	Mosayyapeta	R	12.4	Minor	Panchayat Pucca
31	14160	Mosayyapeta	R	12.5	Minor	School
32	14260	Mosayyapeta	R	3.5	Major	Nukambica Temple

Table 8-2: Complete Details of affecting CPRs in Anakapalle to Atchuthapuram Road

S.No	CPR Description	Image
1	Shiva Temple: It is a type of Religious CPR in Nagulapally village located on left side at chainage 2.38km at a distance of 12.5m Impact: Minor	
2	Devi perantalu kotalu Temple: It is a type of Religious CPR in Nagulapally village located on right side at chainage 2.4 km at a distance of 11m Impact: Major	
3	Satthamma Temple: It is a type of Religious CPR in Nagulapally village located on left side at chainage 2.645km at a distance of 5m Impact: Major	
4	Sri Kasi Annapurneshwari Temple: It is a type of Religious CPR in Nagulapally village located on right side at chainage 2.79km at a distance of 7.5m. Impact: Major	
5	Hanuman Temple: It is a type of Religious CPR in Nagulapally village located on left side at chainage 2.84km at a distance of 8.5m Impact: Major	
6	Bus stand: It is a type of CPR in Nagulapally village located on left side at chainage 2.850km at a distance of 10m Impact: Major	

7	ZPPS SCHOOL: It is a type of CPR in Nagulapally village located on right side at chainage 2.9 km at a distance of 12m. Impact: Major	
8	Vinayaka Temple: It is a type of Religious CPR in Nagulapally village located on right side at chainage 3.12km at a distance of 6.5m. Impact: Major	
9	Pydimamba & Vinayaka Temple: It is a type of Religious CPR in Ompolu village located on right side at chainage 3.82km at a distance of 6.5m. Impact: Major	
10	Pasara maramma thalli temple: It is a type of Religious CPR in Munagapaka village located on left side at chainage 4.74km at a distance of 8m. Impact: Major	
11	Church Compound: It is a type of Religious CPR in Munagapaka village located on right side at chainage 5.530km at a distance of 13.5m. Impact: Minor	
12	Musili Thalli Temple: It is a type of Religious CPR in Munagapaka village located on left side at chainage 5.590km at a distance of 6m Impact: Major	
13	Hanuman Temple: It is a type of Religious CPR in Munagapaka village located on right side at chainage 5.635km at a distance of 8.5m. Impact: Major	

14	Bus stand: It is a type of CPR in Munagapaka village located on right side at chainage 5.650km at a distance of 5.7m. Impact: Major	
15	Pump House: It is a type of CPR in Munagapaka village located on right side at chainage 5.680km at a distance of 13m. Impact: Minor	
16	Shivalayam Temple: It is a type of Religious CPR in Munagapaka village located on right side at chainage 5.7km at a distance of 5m. Impact: Major	
17	Maridamma Thalli Temple: It is a type of Religious CPR in Munagapaka village located on right side at chainage 6.240km at a distance of 11.3m Impact: Major	
18	Maridamma Thalli Temple: It is a type of Religious CPR in Munagapaka village located on right side at chainage 7.430km at a distance of 6m. Impact: Major	
19	Anjaneya Swami Temple: It is a type of Religious CPR in Munagapaka village located on left side at chainage 7.720km at a distance of 7m. Impact: Major	
20	Mutyalamma Temple: It is a type of Religious CPR in Munagapaka village located on right side at chainage 7.815km at a distance of 3.2m. Impact: Major	
21	Mamidi Mahalakshmi Temple: It is a type of Religious CPR in Munagapaka village located on right side at chainage 8.260km at a distance of 5m. Impact: Major	

22	Yellaramma Temple: It is a type of Religious CPR in Kondlakarla village located on left side at chainage 8.635km at a distance of 8.5m. Impact: Major	
23	Vinayaka Temple: It is a type of Religious CPR in Kondlakarla village located on left side at chainage 8.640km at a distance of 13.5m. Impact: Minor	
24	Hanuman Temple: It is a type of Religious CPR in Kondlakarla village located on right side at chainage 8.670km at a distance of 6.5m Impact: Major	
25	Chodambika Temple: It is a type of Religious CPR in Chodapalli village located on right side at chainage 12.955km at a distance of 3m Impact: Major	
26	Paradesimamba pucca: It is a type of CPR in Chodapalli village located on right side at chainage 13.410km at a distance of 8m Impact: Major	
27	Paradesimamba pucca: It is a type of CPR in Chodapalli village located on left side at chainage 13.415km at a distance of 10.8m. Impact: Major	
28	Paradesimamba semi pucca: It is a type of CPR in Chodapalli village located on right side at chainage 13.705km at a distance of 3.8m. Impact: Major	
29	School Compound walls & Toilets: It is a type of school CPR in Mosayyapeta village located on right side at chainage 14.105km at a distance of 12.3m Impact: Minor	
30	Panchayat Pucca: It is a type of Govt office CPR in Mosayyapeta village located on right side at chainage 14.150km at a distance of 12.4m Impact: Minor	
31	School: It is a type of CPR in Mosayyapeta village located on right side at chainage 14.160km at a distance of 12.5m Impact: Minor	

32	Nukambika Temple: It is a type of Religious CPR in Mosayyapeta village located on right side at chainage 14.260km at a distance of 3.5m Impact: Major	
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9 Findings and Recommendations

9.1 Findings: Positive and Negative Impacts

The following are the Consultant's initial findings in regard to likely positive and negative impacts.

9.2 Positive Impacts:

The improved road: will reduce travel times, fuel consumption and emissions from base traffic volumes.

Drainage conditions will be considerably improved: due to the provision of improved side drains, culverts and causeways Economic development and access will be stimulated.

9.3 Negative Impacts:

- Minor deteriorations in the present levels of air, water and noise quality may be expected during construction - but this should be short-term and localised in order to minimise the impacts, the mitigation measures recommended in previous Chapter should be followed during detailed design and construction.
- Construction will require the uprooting of a large number of trees (about. 1737 Nos). This will negatively affect the ecological balance of the area, the impact is, however, expected to be short term, localised and temporary in nature.
- Uprooting of trees will also enhance soil erosion and Disposal of waste may lead to destabilisation of downstream water courses and landscapes.

9.4 Significant Impacts:

Presently, the only "significant" impacts identified are:

- Cutting /clearing of trees and vegetation, due to widening of road
- Degradation of down-stream water quality during construction
- Occasional disturbance from noise generated during construction of Project road leading to disruption in livestock, and annoyance to local residents.
- Occasional high concentrations of air-borne dust during construction of Project road, resulting in deposition and some damage to vegetation, crops and water quality of ponds and irrigation channels.

To minimise the above negative impacts: the following suggested mitigation measures should be incorporated during detailed design and construction

- Stockpiled sand and soil should be slightly wetted before loading, particularly in windy conditions, vehicles transporting sand and soil should be covered with a tarpaulin.
- Blasting and the operation of heavy equipment should be performed only in daylight hours.
- Stone crushing plants should be located away from population centres, drinking water intakes, cultivable lands and sensitive ecosystems - stone crushing equipment should be fitted with approved dust control devices and operated in accordance with manufacturer's specifications.
- Hazardous materials should not be stored near surface water - all used lubricants and oils should be collected and recycled or disposed of the site and impervious layer/sheeting should be placed under hazardous material storage areas to collect and retain leaks and spills.
- Contaminated runoff from storage areas should be captured in ditches with an oil trap at the outlet. Contaminated and worn plastic sheeting should be packed into drums and disposed of the site.
- Use of fuel wood for heating bitumen should be discouraged, where heating is required, bitumen heaters should be used, fuelled by kerosene, diesel or gas – no bituminous material should be discharged into side drains

9.5 Recommendations

Contracts for Construction and the Bills of Quantities should incorporate the requirements for all the earlier-referred to environmentally-recommended working practices and for the Consultant's suggested measures designed to mitigate long-term post construction adverse impacts.

The following are deemed to be the most important initial recommendations designed to make the project environmentally sound and sustainable.

- The Contractors should comply with all environmental mitigation measures specified in this IEE report.
- First priority should be to reuse excess material/debris generated during construction works.
- Discarded materials that cannot be used in construction and fill should, if suitable, be used for bio-engineering measures.
- All other excess material should be disposed of in locations or landfills that will not promote instability and result in destruction of property, vegetation, irrigation and drinking water supply systems.

- Extreme care should be taken to avoid disposal near water-bodies or in areas that will inconvenience or deprive local residents of their livelihood.
- The Contractors should consult with the Engineer-in-Charge before locating project offices, sheds and construction plants.
- Camps should not be located near settlements (or) irrigation ponds, nor to negatively impact local resident's access to drinking water.
- Camps should be operated within a self-sufficient infrastructure and no trees should be cut for wood fuel.
- The implementation unit should ensure that the contractors identify environmentally safe locations for siting construction camps and other plants, machinery, vehicles and equipment, as well as locations for storage and disposal of wastes, both from the construction camps and from the sites.
- Environmental and Social Management Cell of implementation unit should have capacity to address environmental and social measures related to implementation of proposed project.
- The implementation unit should ensure that the regulatory permissions required for the construction equipment, vehicles and machinery have been obtained by the Contractors and are valid at all times during the execution of the project.
- The implementation unit should ensure that pollution control equipment is installed on all equipment and functions properly by the Contractor.
- During haul operations, measures should be enforced to ensure environmental protection, namely regular sprinkling of water to suppress dust, use of spillage proof vehicles for hauling and regular maintenance of vehicles etc.
- Contractor should be required to maintain construction sites, keep them clean and provide appropriate facilities for the storage of all waste until it is disposed.
- Work forces should be provided with adequate water supply, septic tanks with soak pits, health care facilities and LPG to reduce their dependence on wood fuels.
- The Contractors should ensure that the safety of workers and other site users is not compromised during construction.
- Quarry area should be restricted along the road and close to the RoW.

- Where possible, quarry area should be located away from population centres, streams, drinking water intake, cultivable lands and natural drainage systems.
- No bituminous material should be discharged into side drains.
- Nearby trees, vegetation and private property should be protected during bitumen emulsion spraying work.
- Bitumen drums should be stored in designated locations and not scattered along the road.
- Borrow areas should not be established in ecologically sensitive areas.
- Borrow areas should be located away from the road and settlements facing the road - so as to minimise visual impacts.
- Vehicles delivering loose and fine materials should be covered to reduce spills.
- Particular care/attention should be given to adequate drainage.
- For the construction zones within 500 metres of settlements, noisy operations should be stopped between 22:00 to 06:00 hrs.
- Care should be taken to avoid sacred and religious sites.
- Loss of trees should be compensated by planting trees on available space in consultation with forest department.
- If required, Environmental Management and Monitoring Plan provided in Chapter 5 may be updated suitably during implementation.

10 Conclusions

On the basis of (i) Information available, (ii) Field visits over the entire length of the project road, (iii) Discussions with the Project authorities; (iv) Consultation with local people and various Governmental officials, it has been concluded that:

- All elements of the projects will be beneficial;
- All negative impacts, during and after construction, including those deemed “significant” can be properly mitigated; and
- No comprehensive, broad, diverse or irreversible adverse impacts have been identified.

For all the works, the APRDC will, however, have to obtain required clearances from the Environmental Department and Pollution Control Board.

Attachment 1

Rapid Environmental Assessment (REA) Checklist Anakapalle to Atchutapuram Road

Country/Project Title: INDIA: Visakhapatnam-Chennai Industrial Corridor

Sector Division: Andhra Pradesh Road Development Corporation

Road Section: Samarlakota-Rajanagaram Km 26+400 to Km 56+000

Screening questions	Yes	No	Remarks
A. Project siting Is the project area adjacent to or within any of the following environmentally sensitive areas?			
▪ Cultural heritage site	X		The Archaeological Survey of India protected 'Kumar Bhimeswara Swami Temple' is 1.5 Km away from Samarlakota starting point. No direct impact envisaged.
▪ Protected area		X	
▪ Wetland		X	
▪ Mangrove		X	
▪ Estuarine		X	
▪ Buffer zone of protected area		X	
▪ Special area for protecting biodiversity		X	
B. Potential environmental impacts Will the project cause			

Screening questions	Yes	No	Remarks
Encroachment on historical/cultural areas; disfiguration of landscape by road embankments, cuts, fills, and quarries?	X		The topography of project road is mainly plain. Minor impacts of landscape by road embankments. No encroachment of historical places. However, some temples / shrines exist along the project road which may get impacted. Proper management & redevelopment plan will be required during construction to sustain the quarries.
Encroachment on precious ecology (e.g. Sensitive or protected areas)?		X	
Alteration of surface water hydrology of waterways crossed by roads, resulting in increased sediment in streams affected by increased soil erosion at construction site?		X	Project road does not cross the water-bodies (Roadside ponds/Irrigation tanks) which are protected by earthen bunds.
Deterioration of surface water quality due to silt runoff and sanitary wastes from worker-based camps and chemicals used in construction?		X	Adequate sanitary facilities and drainage in the workers camps will help to avoid this possibility. The construction activity wherein chemicals are used shall follow manufacturer's recommendation to prevent impact on surface water quality.

Screening questions	Yes	No	Remarks
Increased local air pollution due to rock crushing, cutting and filling works, and chemicals from asphalt processing?	X		With appropriate mitigation measures and use of most modern environment friendly equipment's/machine ries air pollution shall be reduced to permissible levels.
Noise and vibration due to blasting and other civil works?	X		Short term minor impact may occur during construction period, suitable mitigation measures will be required to minimize the adverse effects
Dislocation or involuntary resettlement of people?	X		In case of loss of land, compensation for land at replacement cost for land for land, where feasible. In case of severance of land, the landowner will have the option of offering the unviable severed proportion of the land for acquisition. The value of houses, buildings & other immovable properties will be determined by R&B Dept. on basis of SSR as on date without depreciation & the solatium also.

Screening questions	Yes	No	Remarks
Other social concerns relating to inconveniences in living conditions in the project areas that may trigger cases of upper respiratory problems and stress?		X	Imposing of appropriate mitigation measures in contract agreement to keep the air pollution within permissible levels will keep a check on this problem.
Inconvenience due to landslide or erosion?		X	The possibility of landslide and erosion at the quarry area/borrow area may be avoided by implementation of Quarry/Borrow area management plan taking into consideration the slope stability norms. Silt-fence provision shall be made to arrest the erosion, if unavoidable.
Inconvenient environmental condition due to poor sanitation and solid waste disposal in construction camps and work sites, and possible transmission of communicable diseases from workers to local populations?	X		Proper provisions for sanitation, health care and solid waste disposal facilities will be available in the contract documents to avoid such possibility. Workers shall be made aware about communicable diseases.
Creation of temporary breeding habitats for mosquito vectors of disease?		X	Proper waste-water management and house-keeping may avoid such possibility.

Screening questions	Yes	No	Remarks
Accident risks associated with increased vehicular traffic, leading to loss of life?		X	Adoption of suitable traffic signage system at sensitive places/habitation and road safety education to local people will reduce such possibility.
Increased noise and air pollution resulting from traffic volume?		X	Due to improvement in Riding Quality & Comfort in driving due to unidirectional traffic such pollution will be reduced. Mitigation measures along with monitoring plan will be required
Increased risk of water pollution from oil, grease and fuel spills, and other materials from vehicles using the road?	X		Controlled construction activities and proper drainage system with oil/grease trap will reduce this possibility.

After reviewing the answers above the Mission Leader and Environment Specialist agree that the project.

	Should be categorized as an A project.
X	Should be categorized as a B project.
	Should be categorized as a B project in an environmentally sensitive area.
	Should be categorized as a C project.
	Should be categorized as an A/B project because (give reason) .
	Requires additional information for classification.
	Therefore, an Environment Specialist should be involved in the PPTA Fact-finding Mission.

	The Mission Leader should gather additional information during the PPTA Fact-finding Mission.
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PUBLIC CONSULTATION AT MUNAGAPAKA

Venue: The Co-Operative Bank Meeting hall at Munagapaka

Date: 12-10-2015 **Time:** 11.00 AM

Participants:

From the Government Department (GoA.P.): Mr.Subbarao and Bargavi, Assistant Executive Engineers, A.P. Road Development Corporation (APRDC), Anakapalle Division.

From the DPR Consultants: Er.A.Madhava Reddy, Environmental Specialist, Mr.P. Devaraju, Social and R&R Specialist, Mr. Praveen Technical Expert.

Public Participated: Potential Project Affected Persons

The Village heads, Public Representatives, The Residential House Owners, Land owners, Retried Government Employs, Shop Owner's, Housewife's, Widows, Village Youth groups, Agriculture Labours, Businessmen's, Private Employs, Government Employs, Vulnerable People and etc.

The Participants List is in closed with Name of the Person, Nature of Affecting Designation with Mobile numbers with Signature.

The Villagers Participated from the Affected Villages at Munagapaka Venue are Munagapaka Anakapalle, Gangadevipeta, Vummalada, Nagulapally, Ompolu, Haripalem, Thimmarajupeta nearly about 350 Persons attended the Public Consultation and Participated.

The Public Consultation convened by the Munagapaka President T. Ramanababu as the Chair Person of the Meeting.

The APRDC Assistant Executive Engineers, Sri Subbarao and Bargavi Presided as the Chief Guest of the Meeting. The Social and Environmental Experts and Technical Team from Roughton and Satra Consultants participated.

The Village Representatives Vummalada-Surpanch, Nagulapally-Surpanch, Ompolu-Surpanch, Haripalem-Surpanch, B.Prasad and Others Participated.

The Public Consultation started at 11:00 AM with the WEL-COMING Speech. The Executive Engineer narrated the importance of VCIC project corridor to the participants and requested every participants to express their views, opinion, suggestions, and objections regarding the proposed road either social or environmental issues.

The Social Environmental and Technical Team of Roughton and Satra consultants, explained clearly about the project social, environmental issues and nature of affecting of settlements, common property

resources(CPRs), water bodies, trees, and etc., The Technical Expert informed about the technical specification of proposed road such as existing RoW and proposed RoW, Curves improvements and importance of usage of existing ADB Road and the proposed RoW at village sections in order to reduce the affecting settlement and L A to the participants.

The Chair Person requested the participants to respond one by one based on the nature of affecting. The participants positively responded and started questioning, expressing their views, suggestions, alternative practices etc.

The entire program was exclusively video graphed with audio recording, right from the starting of the programme to the disbursing of participants. In this records the participants voice captured and documented in the form of compact disk (CD)s, the local media also covered the programmed in major Telugu newspapers.

Issues Discussed in the Public Consultations are:

1. Brief introduction of the project, importance, implementation and funding agency.
2. Loss of structure like Residential, Commercial, Residential & Commercial, CPRs and others.
3. Loss of Agriculture Lands.
4. Compensation of for the affecting structures and lands.
5. Affecting of CPRs.
6. Resettlements and Rehabilitation sites.
7. Road safety and health (HIV/AIDS).
8. Water bodies affecting.
9. Gender issues.
10. Vulnerable persons,
11. Other issues

Suggestions:

1. The villagers attended from the all affecting villages unanimously as single world resolution they expressed that they are opposing for the expansion of the road. For the purpose of industrial growth/development they also resolved that government can lay the road with in the Right of Way (RoW).They totally opposed not to give a feet of residential or agriculture land for purpose of road.
2. The villagers totally objected and requested the consultant agencies not to mark on structures and also rubbed the marking on their structures and bluntly rejected to co-operative. For further enumeration work requested to stop the work immediately.

ANAKAPALLE – ATCHUTAPURAM ROAD PUBLIC CONSULTATIONS AT MUNAGAPAKA VILLAGE



Section of the Participants during the Consultations at Munagapaka Village



Section of the Participants during the Consultations at Munagapaka Village

CPR's Photos



